Heat Transfer

9|11|21|Saturday

* Heat flows from one place to another mainly by three methods.

1. conduction
2. convection
3. Radiation

conduction: solids > liquids > gases

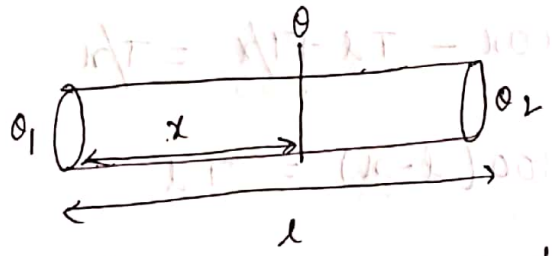
Temp gradient –

$$* \quad \boxed{\frac{-dT}{dx} = -\frac{d\theta}{dx}} \qquad \text{unit} - {}^{\circ}C/m$$

* In steady state temp gradient is constant.
* As we move far temp decreases with time.

Ex!

$(\theta_1 > \theta_2)$



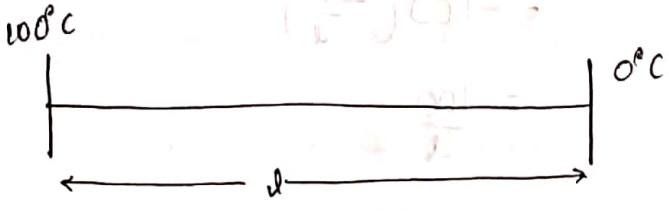
In steady state find temp as function (x).

Sol!

$$\frac{\theta_1 - \theta}{x} = \frac{\theta - \theta_2}{l - x} = \frac{\theta_1 - \theta_2}{l}$$

↳ "θ as a function of x.

Ex!



A stick of cross-section $1cm^2$ and length 1m is

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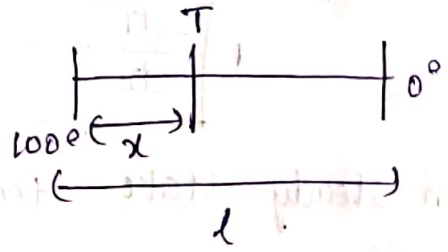
initially at temp of 0°c. It is then kept b/w 2 reserviours of temperatures 100°c & 0°c. Specific heat capacity is 10 J/kg°c and linear mass density is 9 kg/m. Find

(1) Temp gradient along the rod in steady state.

(2) Total heat absorbed by the rod to reach the steady state.

Sol!

In steady state.



(1) Temp gradient = const

$$\frac{100-T}{x} = \frac{T-0}{l-x}$$

$$100l - 100x - Tl + Tx = Tx$$

$$100(l-x) = Tl$$

$$T = 100\left(\frac{l-x}{l}\right)$$

$$\boxed{T = 100\left(1-\frac{x}{l}\right)}$$

$\therefore$ Temp gradient $= -\frac{dT}{dx}$

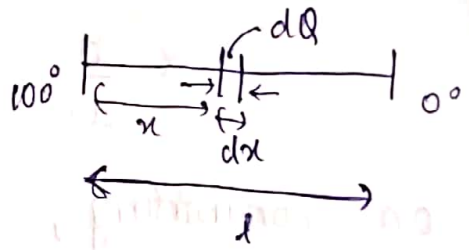
$$= -100\left(-\frac{1}{l}\right)$$

$$= \frac{100}{l}$$

Given $l = 1m \Rightarrow T.g = 100\,°c/m$

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(2) Heat absorbed by the element,



$$dQ = (dm)\,c\,dT$$

$$dQ = \lambda dx\, c\,(T-0)$$

$$dQ = \lambda c\, 100\left(1 - \frac{x}{l}\right) dx$$

$$\int_{x=0}^{x=l} dQ \Rightarrow Q = 100\lambda c \int_{x=0}^{l}\left(1 - \frac{x}{l}\right) dx$$

$$= 100\lambda c \left[x - \frac{x^2}{2l}\right]_0^l$$

$$= 100\lambda c \left[l - \frac{l^2}{2l}\right]$$

$$= 100\lambda c\, l/2$$

$$= 500\lambda c l$$

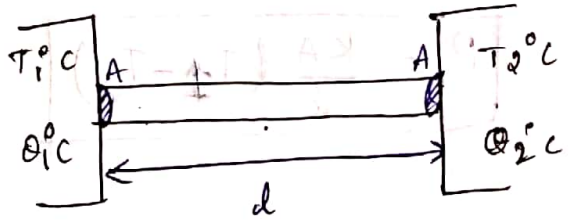
$$= 50 \times 2 \times 10 \times 1$$

$$\boxed{Q = 1000\ J}$$

given:
$\lambda = 2\ kg/m$
$c = 10\ J/kg°C$

Coefficient of thermal conductivity (K):

$(T_1 > T_2)$



* Every solid not allow heat in the same manner.

* Rate of heat flow (power of heat flow):

$$P\left(\frac{dQ}{dT}\right) \propto A$$ — In steady state.

$$P \propto \frac{-dT}{dx}$$

on combining,

$$P \propto -A\frac{dT}{dx}$$

$$\boxed{P = -KA\frac{dT}{dx}} \quad ***$$

where K — coeff of thermal conducity.

Here, $P = -KA\frac{dT}{dx}$

$$\int_0^l dx = \int_{T1}^{T2} -\frac{KA\,dT}{P}$$

$$l = -\frac{KA}{P}(T_2 - T_1)$$

$$l = \frac{KA}{P}(T_1 - T_2)$$

$$\boxed{P = \frac{KA}{l}(T_1 - T_2)}$$

$$\frac{Q}{t} = \frac{KA}{l}(T_1 - T_2)$$

$$\boxed{Q = \frac{KA}{l}(T_1 - T_2)t}$$

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$$\text{eq} \quad K = \frac{Qd}{A(T_1 - T_2)t}$$

$$K = \frac{Q}{At\left(\frac{T_1 - T_2}{d}\right)}$$

* K is said to be rate of heat per unit Area per unit temp gradient.

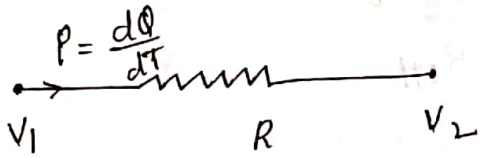
* unit for $K = \frac{J}{m^2 \times s \times {}^{\circ}C} = \frac{J}{m\,s\,{}^{\circ}C}$ (S.I) $Wm^{-1}{}^{\circ}C^{-1}$

* K depends on nature of material but doesn't depends on dimensions of body.

* K ↑ → good conductor of heat

* Silver → K – highest.

*



$$\boxed{P = \frac{V_1 - V_2}{R}}$$

here, $P = \frac{KA(T_1 - T_2)}{d}$

$$P = \frac{T_1 - T_2}{\frac{d}{KA}}$$

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∴ Thermal resistance of body,

$$R_{th} = \frac{l}{KA}$$

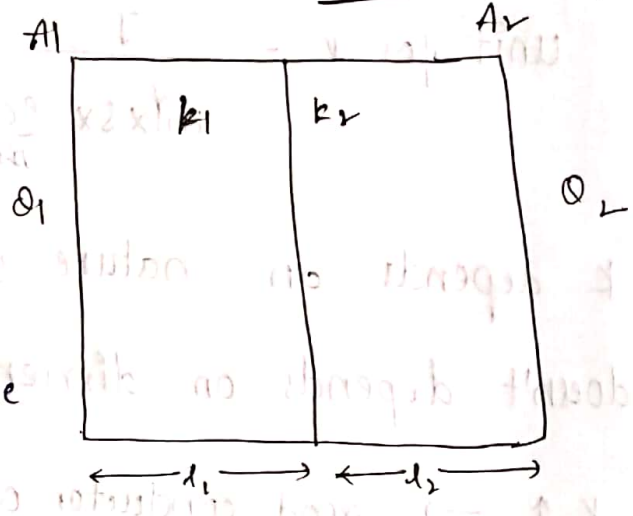
units: $R_{th} = \frac{\Delta T}{P} \longrightarrow {}^{\circ}C/\text{watt}$

* R_{th} depends on 1) nature of material (K)
2) dimensions of body

11/11/24 Monday

Slabs in series:

[here $(\theta_1 > \theta_2)$]



① In steady state, rate of heat flow is same through all the slabs,

we have $\Delta\theta = PR$

$$P = \frac{\theta_1 - \theta}{R_1} = \frac{\theta - \theta_2}{R_2} = \frac{\theta_1 - \theta_2}{R_{eff}}$$

$$R_1 = \frac{l_1}{K_1 A_1}, \quad R_2 = \frac{l_2}{K_2 A_2}$$

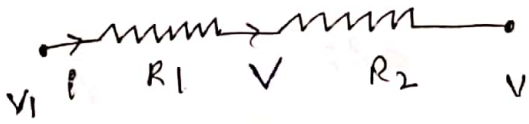
$$\left(\frac{\theta_1 - \theta}{l_1}\right) K_1 A_1 = \left(\frac{\theta - \theta_2}{l_2}\right) K_2 A_2$$

$$\theta_1 K_1 A_1 l_2 - \theta K_1 A_1 l_2 = \theta K_2 A_2 l_1 - \theta_2 K_2 A_2 l_1$$

$$\theta = \frac{K_1 A_1 \theta_1 l_2 + K_2 A_2 \theta_2 l_1}{K_1 A_1 l_2 + K_2 A_2 l_1}$$

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* In series,



$$i = \frac{V_1 - V}{R_1} = \frac{V - V_2}{R_2} = \frac{V_1 - V_2}{R_{eff}} \quad **$$

i same through both the coires.

we have

$$V_1 - V_2 = (V_1 - V) + (V - V_2)$$

$$i R_{eff} = i R_1 + i R_2$$

$$R_{eff} = R_1 + R_2 \quad **$$

$$i = \frac{V_1 - V_2}{R_1 + R_2} \quad \& \quad V_1 - V = iR_1$$

$$V - V_2 = iR_2$$

$$\frac{V_1 - V}{V - V_2} = \frac{R_1}{R_2}$$

junction potential V can be found.

* If $A_1 = A_2$

$$\theta = \frac{K_1 \theta_1 l_2 + K_2 \theta_2 l_1}{K_1 l_2 + K_2 l_1}$$

Again,

$$\theta_1 - \theta_2 = (\theta_1 - \theta) + (\theta - \theta_2)$$

$$R_{eff} = R_1 + R_2$$

$$R_{eff} = R_1 + R_2$$

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In general,

$$R_S = R_1 + R_2 + \dots = \sum R_i$$

for continuous elements connected in series,

$$\boxed{R_S = \int dR} \quad *$$

Effective thermal conductivity:

$$R_S = R_1 + R_2 + \dots$$

$$\frac{l_1}{K_S A_1} + \frac{l_2}{K_S A_2} + \dots = \frac{l_1}{K_1 A_1} + \frac{l_2}{K_2 A_2} + \dots$$

$$\boxed{K_S = \frac{\frac{l_1}{A_1} + \frac{l_2}{A_2} + \dots}{\frac{l_1}{K_1 A_1} + \frac{l_2}{K_2 A_2} + \dots}} \quad *$$

In general, $P = -KA\frac{d\theta}{dx}$

$$\boxed{\frac{d\theta}{dx} = -\frac{P}{KA}} \quad *$$

In series combination, P = same through all slabs in steady state,

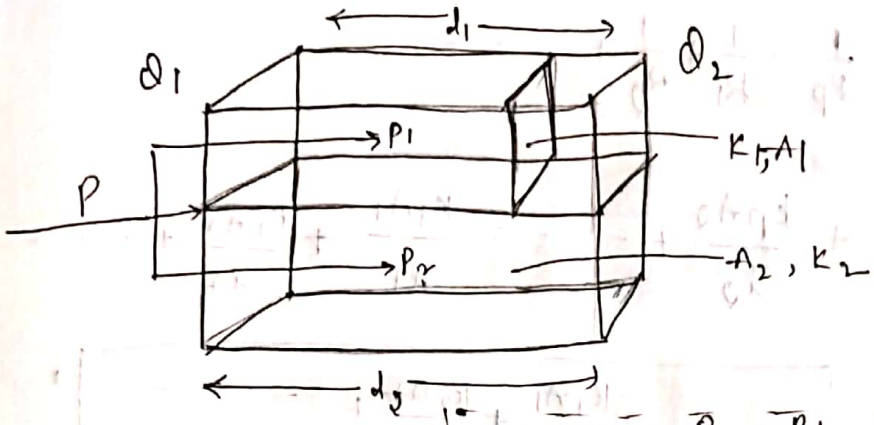
$$\boxed{\frac{-d\theta}{dx} \propto \frac{1}{KA}} \quad **$$

For same temp gradient, $\frac{1}{KA}$ = const

$$\boxed{K_1 A_1 = K_2 A_2} \quad **$$

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Slabs in Parallel Combination:



Here,

① $\Delta\theta = \theta_1 - \theta = \theta - \theta_2$ is same across each slab.

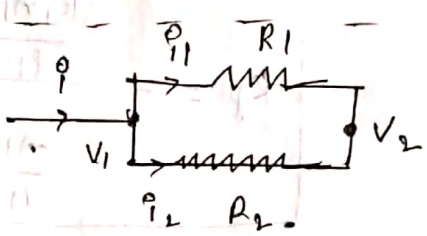
② $P = P_1 + P_2$ (net rate of heat flow)

$$\frac{\Delta\theta}{R_{parallel}} = \frac{\Delta\theta}{R_1} + \frac{\Delta\theta}{R_2}$$

$$\boxed{\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}} \quad *$$

where, $R_1 = \frac{l_1}{K_1 A_1}$

$R_2 = \frac{l_2}{K_2 A_2}$.



$i = i_1 + i_2$

here, $\Delta V = V_1 - V_2$ is same across each resistor.

i.e, $i_1 = \frac{V_1 - V_2}{R_1}$, $i_2 = \frac{V_1 - V_2}{R_2}$

$i = \frac{V_1 - V_2}{R_{eff}}$

$$\boxed{\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2}}$$

In general,

$$\boxed{\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \text{- - - - - } = \sum \frac{1}{R_i}}$$

In integration form,

$$\boxed{\frac{1}{R_p} = \int \frac{1}{dR}}$$

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Effective thermal conductivity:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$$

$$\frac{K_p A_1}{l_1} + \frac{K_p A_2}{l_2} + \cdots = \frac{K_1 A_1}{d_1} + \frac{K_2 A_2}{d_2} + \cdots$$

$$K_p = \frac{\frac{K_1 A_1}{l_1} + \frac{K_2 A_2}{d_2} + \cdots}{\frac{A_1}{d_1} + \frac{A_2}{d_2} + \cdots} \quad *$$

Again, we have,

$$P = -KA\frac{d\theta}{dx}$$

$$\frac{-d\theta}{dx} = \frac{P}{KA} \quad *$$

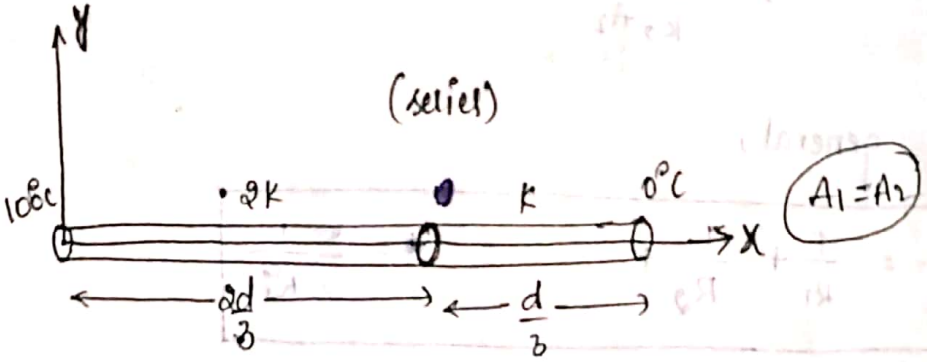
To have same temp gradient,

$$\frac{P}{KA} = \text{const}$$

$$\frac{P_1}{K_1 A_1} = \frac{P_2}{K_2 A_2} \quad *$$

Ex:

①



Sol:

In steady state,

$$\theta = \frac{\theta_1 K_1 A_1 l_2 + \theta_2 K_2 A_2 l_1}{K_1 A_1 l_2 + K_2 A_2 l_1}$$

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$$\theta = \frac{100\left(\frac{d}{3}\right)(2K) + 0 \times (K)\frac{2d}{3}}{2K\left(\frac{d}{3}\right) + K\left(\frac{2d}{3}\right)}$$

$$= \frac{\frac{200dK}{3}}{\frac{4dK}{3}} = 50$$

$$\boxed{\theta = 50^\circ}$$

② Rate of heat through the combination,

$$P = \frac{100-\theta}{R_1} = \frac{\theta-0}{R_2} = \frac{100-0}{R_s}$$

$$R_1 = \frac{l_1}{K_1 A_1}, \quad R_2 = \frac{l_2}{K_2 A_2}$$

$$P = \frac{100-50}{\frac{\frac{2d}{3}}{(2K)A}} = \frac{300KA}{2d} = \frac{150KA}{d}$$

↓

as same through both the rods.

③ Equivalent thermal resistance,

$$R_s = R_1 + R_2$$

$$= \frac{\frac{2d}{3}}{2KA} + \frac{\frac{d}{3}}{KA} = \frac{2d}{3KA}$$

④ Effective thermal conductivity,

$$\frac{l_1}{K_s A_1} + \frac{l_2}{K_s A_2} = \frac{l_1}{K_1 A_1} + \frac{l_2}{K_2 A_2}$$

$$\frac{1}{K_s} = \frac{\frac{\frac{2d}{3}}{2K} + \frac{\frac{d}{3}}{K}}{\frac{2d}{3} + \frac{d}{3}} = \frac{\frac{2d}{3K}}{d} = \frac{2}{3K}$$

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$$\boxed{K_S = \frac{3K}{2}}$$

Graphs-

we have,

$$P = -KA\frac{d\theta}{dx}$$

$$\frac{d\theta}{dx} = -\frac{P}{KA}$$

If K, A are const,

$$\frac{d\theta}{dx} = \text{const}$$

i.e, slope of θ vs x is constant.

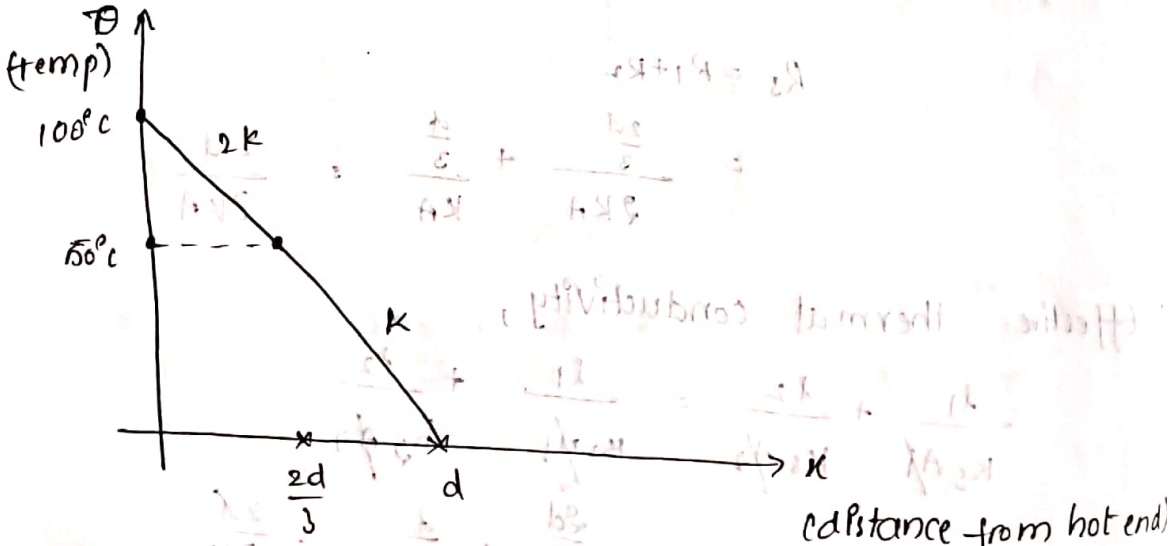
∴ we get st. lines.

now, for ist rod,

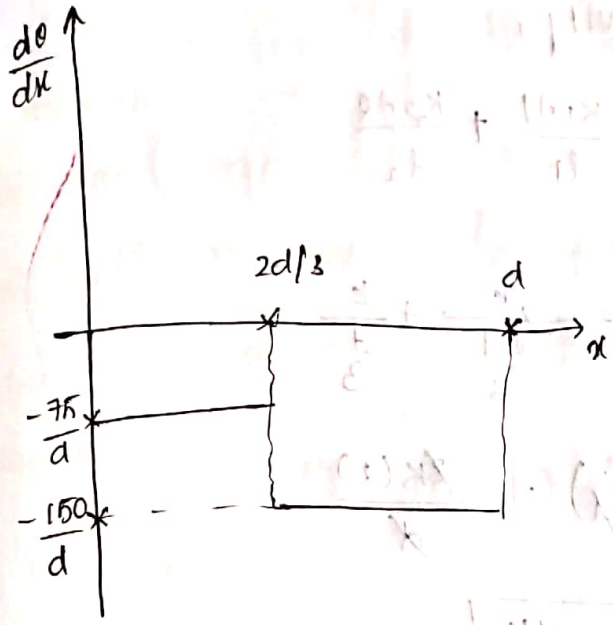
$$\frac{-d\theta}{dx} = \frac{P}{K_1 A} = \frac{\frac{150KA}{d}}{2KA} = \frac{75}{d}$$

for second rod,

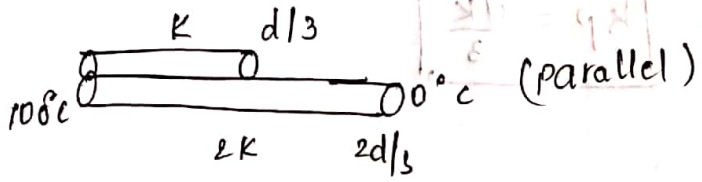
$$\frac{-d\theta}{dx} = \frac{P}{K_2 A} = \frac{\frac{150KA}{d}}{KA} = \frac{150}{d}$$



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Ex!



(parallel)

Sol! rate of heat flow through 1st rod,

$$P_1 = \frac{\Delta\theta}{R_1} = \frac{100-0}{\frac{d}{3}} = \frac{300KA}{d}$$

through 2nd rod,

$$P_2 = \frac{\Delta\theta}{R_2} = \frac{\frac{100-0}{2d}}{\frac{3}{2KA}} = \frac{300KA}{d}$$

net heat flowing through the combination,

$$P = P_1 + P_2 = \frac{600KA}{d}$$

effective thermal resistance,

$$\frac{1}{R_p} = \frac{1}{P_1} + \frac{1}{P_2} \Rightarrow \frac{d}{R_p} = \frac{2d}{300KA}$$

$$R_p = \frac{150KA}{d}$$

$$\frac{1}{R_p} = \frac{3KA}{d} + \frac{3KA}{d} \Rightarrow \boxed{R_p = \frac{d}{6KA}}$$

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Effective thermal conductivity,

$$\frac{K_p A_1}{d_1} + \frac{K_p A_2}{d_2} = \frac{K_1 A_1}{l_1} + \frac{K_2 A_2}{l_2}$$

$$\frac{K_p}{\frac{2d}{3}} + \frac{K_p}{\frac{d}{3}} = \frac{2K}{\frac{2d}{3}} + \frac{K}{\frac{d}{3}}$$

$$3K_p\left(\frac{3}{2d}\right) = \frac{3K(2)}{d}$$

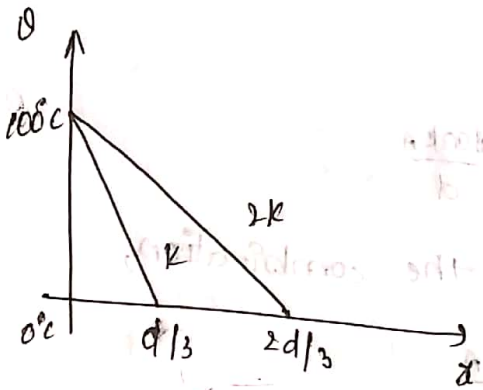
$$\boxed{K_p = \frac{4K}{3}}$$

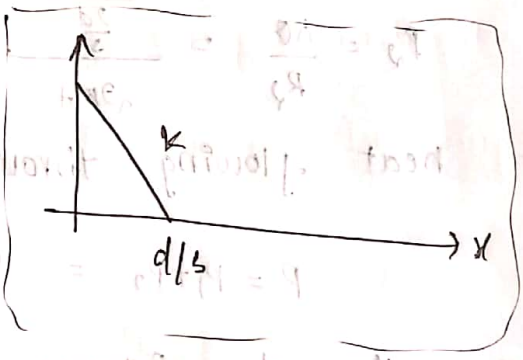
Graphs:

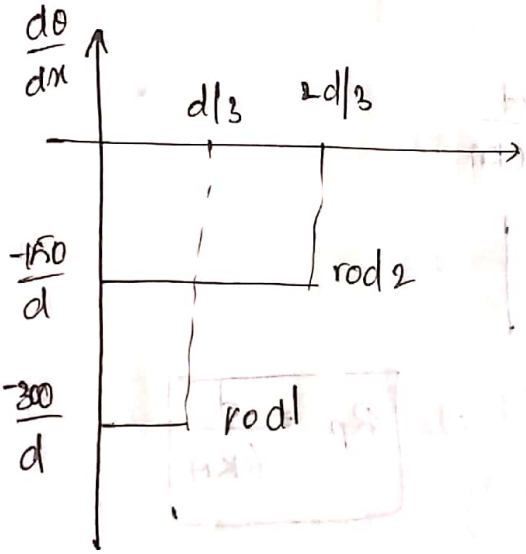
$$\frac{d\theta}{dx} = \frac{-P}{KA}$$

for 1st rod, $\left(\frac{d\theta}{dx}\right)_1 = \frac{-P_1}{K_1 A_1} = \frac{-300KA}{d \times 2KA} = \frac{-150}{d}$

for 2nd rod, $\left(\frac{d\theta}{dx}\right)_2 = \frac{-P_2}{K_2 A_2} = \frac{-300}{d}$

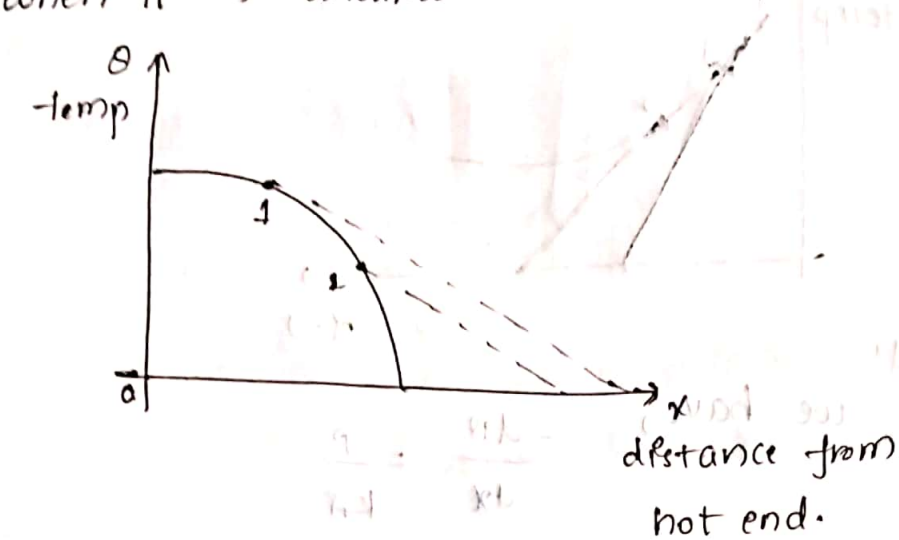






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Ex: The ends of a long homogeneous bar are maintained at different temps and there is no loss of heat from the sides of bar due to conduction (or) radiation. The graph of the temp against distance of the bar when it is attained in steady state is as shown.



Sol:

we have, $-\frac{d\theta}{dx} = \frac{P}{KA}$

for homogeneous rod, K = const

in steady state, P = const

$$\boxed{\text{slope} \propto \frac{1}{A}}$$

$\frac{d\theta}{dx}$ is not const, i.e, temp gradient is not same along the rod.

clearly it is increasing.

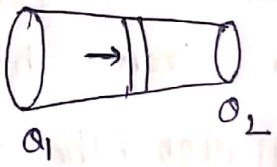
Since slope of 1 < slope of 2.

$$\left|\frac{-d\theta}{dx}\right|_1 < \left|\frac{-d\theta}{dx}\right|_2$$

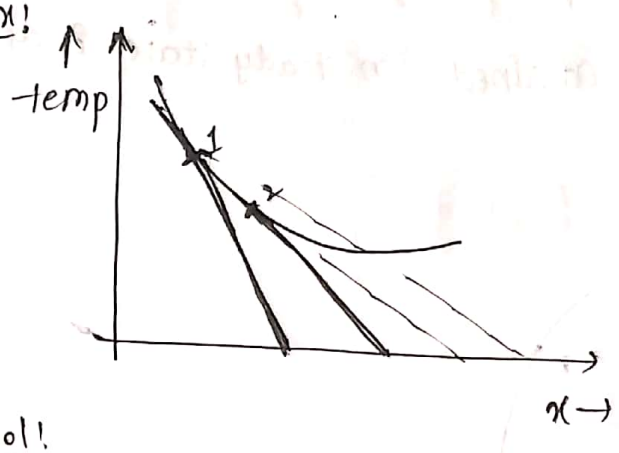
$$\frac{1}{A_1} < \frac{1}{A_2} \Rightarrow \boxed{A_1 > A_2}$$

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Area of cross-section is not uniform.



Ex!



Sol!

we have, $\frac{-d\theta}{dx} = \frac{P}{KA}$

for homogeneous rod, $K = \text{const}$

In steady state, $P = \text{const}$

slope $\propto \frac{1}{A}$

$\frac{d\theta}{dx}$ is not const, i.e, temp. gradient is not same.

along the rod,

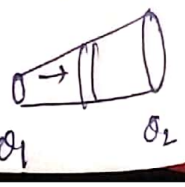
clearly it is decreasing

$$\left|\frac{-d\theta}{dx}\right|_1 > \left|\frac{-d\theta}{dx}\right|_2$$

$$\frac{1}{A_1} > \frac{1}{A_2}$$

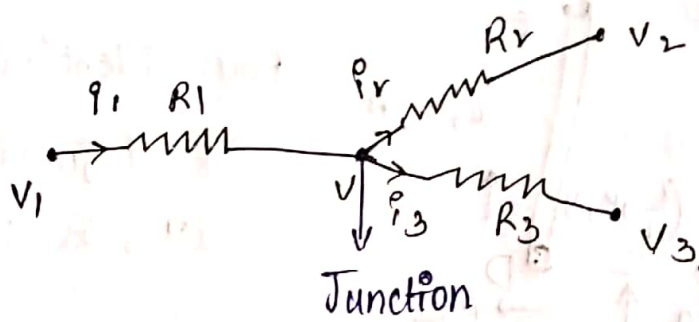
$$\boxed{A_2 > A_1}$$

Area of cross-section is not uniform.



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Concept:



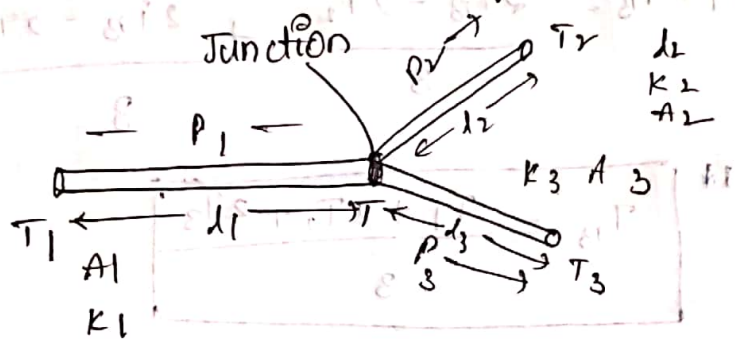
From junction law,

$$i_1 = i_2 + i_3$$

$$\frac{V_1 - V}{R_1} = \frac{V - V_2}{R_2} + \frac{V - V_3}{R_3}$$

after solving, we get V = junctionpotential.

*



$$P_1 = P_2 + P_3$$

$$\frac{\Delta T_1}{R_1} = \frac{\Delta T_2}{R_2} + \frac{\Delta T_3}{R_3}$$

$$\boxed{\frac{T_1 - T}{R_1} = \frac{T - T_2}{R_2} + \frac{T - T_3}{R_3}}$$

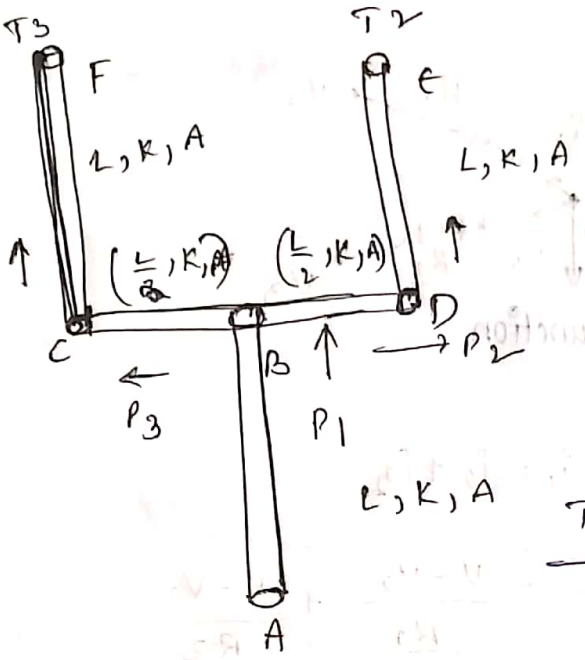
$$R_1 = \frac{l_1}{K_1 A_1}$$

$$R_2 = \frac{l_2}{K_2 A_2}$$

$$R_3 = \frac{l_3}{K_3 A_3}$$

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Ex!



Four identical rods.

Find junction temp

T_B is steady state?

$$P_1 = P_2 + P_3$$

$$\frac{T_1 - T_B}{\frac{L}{KA}} = \frac{T_B - T_2}{\frac{3L}{2KA}} + \frac{T_B - T_3}{\frac{3L}{2KA}}$$

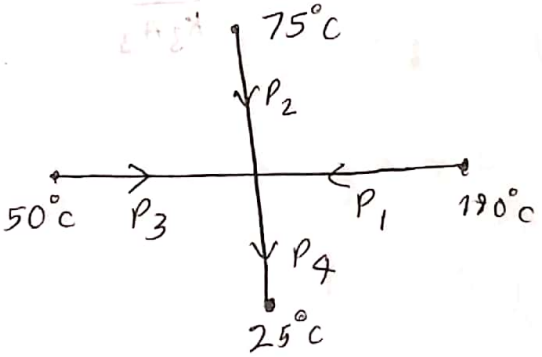
$$T_1 - T_B = \frac{2T_B - 2T_2}{3} + \frac{2T_B - 2T_3}{3}$$

$$\boxed{T_B = \frac{T_1 + 2T_2 + 2T_3}{3}}$$

$$2T_1 - 3T_B = 4T_B - 2T_2 - 2T_3$$

$$\boxed{T_B = \frac{3T_1 + 2T_2 + 3T_3}{7}}$$

Ex! Two identical rods are joined at their middle points, the ends are maintained at const temp as shown. Find the temp of junction.



A) $P_1 + P_2 + P_3 = P_4$

$180 - T + 75 - T + 50 - T = T - 25$

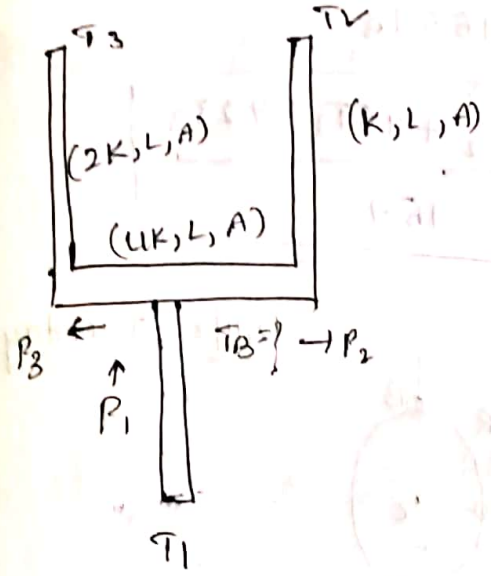
$305 + 25 = 4T$

$T = \frac{330}{4}$

$\boxed{T = 82.5°C}$

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Ex1



$$\frac{T_1 - T_B}{\frac{K}{LA}} = \frac{T_B - T_2}{\frac{4K}{\frac{L}{2}A} + \frac{K}{LA}} + \frac{T_B - T_3}{\frac{4K}{\frac{LA}{2}} + \frac{2K}{LA}}$$

$P_1 = P_2 + P_3$

$$\frac{T_1 - T_B}{\frac{L}{KA}} = \frac{T_B - T_2}{\frac{\frac{L}{2}}{4KA} + \frac{L}{KA}} + \frac{T_B - T_3}{\frac{\frac{L}{2}}{4KA} + \frac{L}{2KA}}$$

$$\frac{T_1 - T_B}{\frac{L}{KA}} = \frac{T_B - T_2}{\frac{9L}{8KA}} + \frac{T_B - T_2}{\frac{5L}{8KA}}$$

$R_{eff_1} = R_1 + R_2$

$= \frac{L}{KA} + \frac{L}{8KA}$

$= \frac{9L}{8KA}$

$R_{eff_2} = R_1 + R_2$

$= \frac{L}{2KA} + \frac{L}{8KA}$

$= \frac{5}{8KA}$

$$T_1 - T_B = \frac{8(T_B - T_2)}{9} + \frac{8(T_B - T_3)}{5}$$

$$45(T_1 - T_B) = 40(T_B - T_2) + 72(T_B - T_3)$$

$$45T_1 - 45T_B = 40T_B - 40T_2 + 72T_B - 72T_3$$

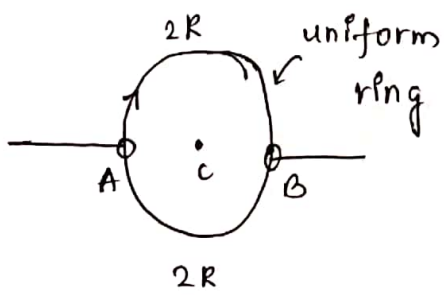
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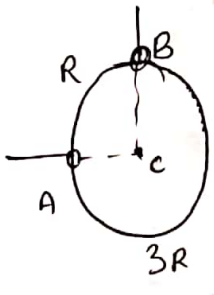
$$45T_1 + 40T_2 + 72T_3 = 157T_B$$

$$\boxed{T_B = \frac{40T_2 + 45T_1 + 72T_3}{157}}$$

12/11/24/Tuesday

Ex1





when $\angle ACB = 180°$, rate of heat flow from A to B is 1.2W. Find rate of heat flow from A to B when $\angle ACB = 90°$.

Sol:

$$P = \frac{\Delta T}{R_i}$$

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R_i}$$

$$R_i = R$$

$$P = \frac{T_B - T_A}{R_i}$$

$$P = \frac{T_B - T_A}{R} = (1.2) \quad -①$$

case ②:

$$P_1 = \frac{T_B - T_{A_1}}{R'_2}$$

$$\frac{1}{R'_1} = \frac{1}{R} + \frac{1}{3R}.$$

$$\frac{1}{R'_1} = \frac{4}{3R}$$

$$P_1 = \frac{4}{3R}(T_B - T_A)$$

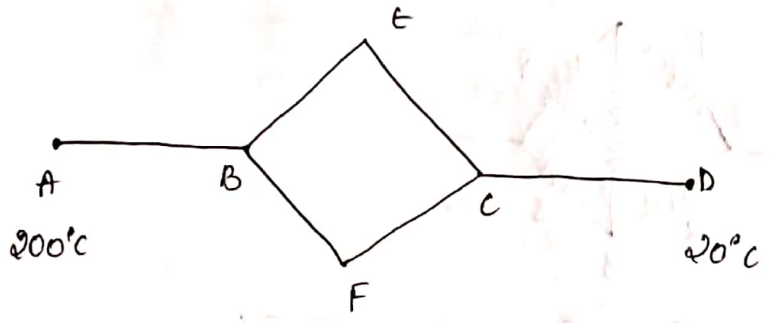
$$= \frac{4}{3}(1.2)$$

$$= 4(0.4)$$

$$\boxed{P_1 = 1.6W}$$

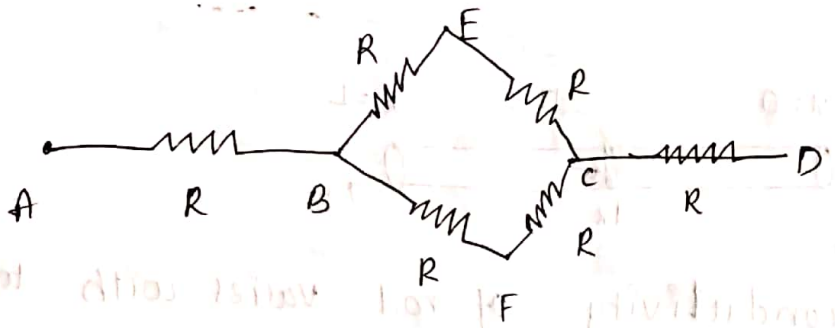
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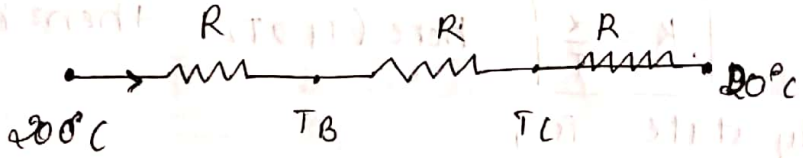
Ex! six identical rods.



In steady state, find T_B, T_C, T_E & T_F.

Sol!





In steady state,

$$P = \frac{200-T_B}{R} = \frac{T_B-T_C}{R} = \frac{T_C-20}{R} = \frac{200-20}{3R}$$

$$\frac{200-T_B}{R} = \frac{60}{3R}$$

$$T_B = 140°C$$

$$\frac{T_C-20}{R} = \frac{60}{3R}$$

$$T_C = 80°C$$

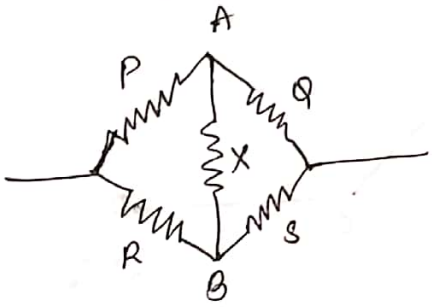
$$\frac{P}{2} = \frac{T_B-T_E}{R}$$

$$\frac{30}{R} = \frac{140-T_E}{R}$$

$$T_E = 110°C = T_F$$

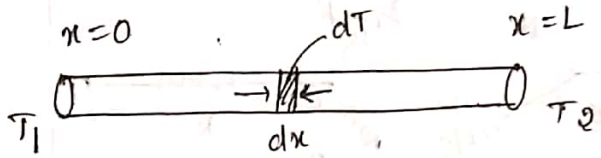
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* Wheatstone Bridge:



If $\frac{P}{Q} = \frac{R}{S}$ then $T_A = T_B$

Ex:



Thermal conductivity of rod varies with temp as

$K = \frac{\alpha}{T}$ here $(T_1 > T_2)$ then heat flow in steady state is,

$$P = -KA \frac{dT}{dx}$$

$$P = -\frac{\alpha}{T} A \frac{dT}{dx}$$

$$\int_0^L P \cdot dx = -\alpha A \int_{T_1}^{T_2} \frac{dT}{T}$$

$$PL = -\alpha A \ln(T_2/T_1)$$

$$P = \frac{\alpha A}{L} \ln(T_1/T_2) \quad *$$

* Thermal resistance of rod,

$$R_{th} = \frac{\Delta T}{P} = \frac{T_1 - T_2}{\frac{\alpha A}{L} \ln(T_1/T_2)}$$

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$$R_{th} = \frac{(T_1 - T_2)L}{\alpha A \ln(T_1/T_2)}$$

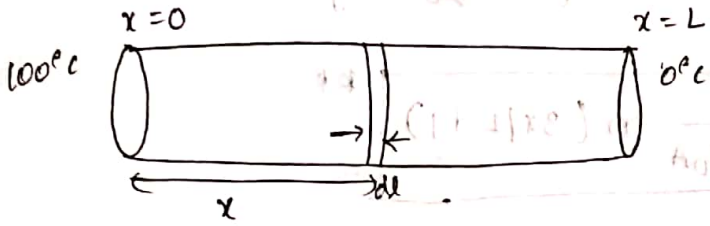
* effective thermal conductivity of rod,

$$R_{th} = \frac{L}{k_{eff} A}$$

$$\frac{(T_1 - T_2)\cancel{L}}{\alpha \cancel{A} \ln(T_1/T_2)} = \frac{\cancel{L}}{k_{eff}\cancel{A}}$$

$$k_{eff} = \frac{\alpha \ln(T_1/T_2)}{T_1 - T_2}$$

Ex!



Here thermal conductivity of rod increases linearly from K_0 to $4K_0$ (from hot end) then,

we have

$$k = ax + b$$

at $x = 0$ $k = K_0$

$$b = K_0$$

at $x = L$

$k = 4K_0$

$4K_0 = aL + K_0$

$$\frac{3K_0}{L} = a$$

$$\Rightarrow k = \frac{3K_0}{L}x + K_0$$

* Thermal resistance,

$$R = \int dR$$

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b/w $x=0$ & $x=x$;

$$R_1 = \int_{x=0}^{x=x} \frac{dx}{kA}$$

$$R_1 = \int_{x=0}^{x=x} \frac{dx}{\left(\frac{3K_0}{L}x + K_0\right)A}$$

$$R_1 = \frac{L}{3K_0A}\left[\ln\left(\frac{3K_0}{L}x + K_0\right)\right]_0^x$$

$$R_1 = \frac{L}{3K_0A} \ln\left[\frac{\left(\frac{3K_0}{L}x + K_0\right)}{K_0}\right]$$

$$\boxed{R_1 = \frac{L}{3K_0A} \ln(3x/L + 1)} \quad **$$

* Total Thermal resistance of rod,

$$R_{th} = \int_{x=0}^{x=L} dR$$

$$= \frac{L}{3K_0A} \ln(3L/L + 1)$$

$$R_{th} = \frac{2L \ln 2}{3K_0A}$$

$$\boxed{R_{th} = \frac{2\ln 2}{3}\frac{L}{K_0A}}$$

In steady state,

$P = \text{const}$

$$\Rightarrow \frac{100-T}{R_1} = \frac{100-0}{R_{th}} = \frac{T-0}{R_2} \quad [R_1 + R_2 = R_{th}]$$

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~~100-T~~ $\quad \frac{100-T}{R_1} = \frac{100}{R_{th}}$

~~R1~~

$$\boxed{T = 100\left(1 - \frac{R_1}{R_{th}}\right)}$$

$$\boxed{T = 100\left(1 - \frac{\ln(3x/L+1)}{2\ln 2}\right)} \quad \text{(from hot end)}$$

$$T = \frac{50}{100}\left(\frac{\ln 4 - \ln(3x/L+1)}{2\ln 2}\right)$$

$$T = \frac{50}{\ln 2}\left(\ln\left[\frac{4}{\left(\frac{3x}{L}+1\right)}\right]\right)$$

* Rate of flow of heat,

$$P = \frac{\Delta T}{R_{th}} = \frac{100-50}{\frac{2\ln 2}{3}\frac{L}{K_0 A}}$$

$$\boxed{P = \frac{150}{\ln 2}\left(\frac{K_0 A}{L}\right)}$$

* Effective thermal conductivity,

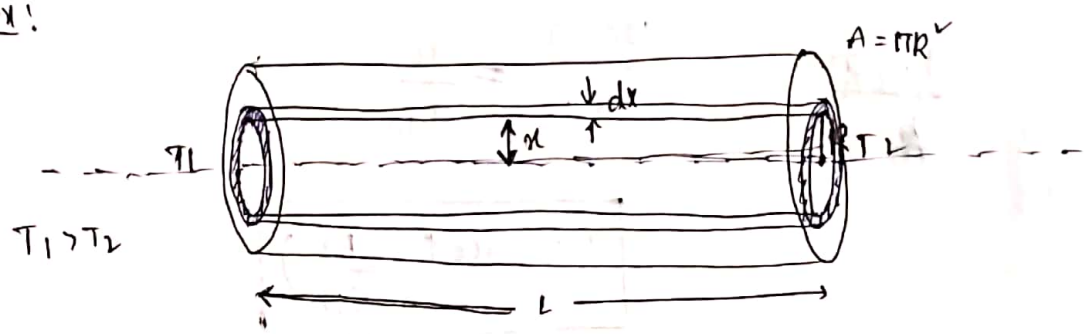
$$R_{th} = \frac{L}{K_{eff} A}$$

$$\frac{2\ln 2}{3} \times \frac{L}{K_0 A} = \frac{L}{K_{eff} A}$$

$$\boxed{K_{eff} = \frac{3K_0}{2\ln 2}}$$

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Ex:



Thermal conductivity of rod * varies with distance 'x' from axis of rod** i.e, $K = K_1 x + K_2$ then find

1) thermal resistance of rod

$$\frac{1}{R_{th}} = \int \frac{1}{dR} \quad \text{(shells are in parallel)}$$

$$\frac{1}{R_{th}} = \int_{x=0}^{x=R} \frac{K\,dA}{l}$$

$$\frac{1}{R_{th}} = \int_{x=0}^{x=R} \frac{(K_1 x + K_2)(2\pi x)\,dx}{l}$$

$$\frac{1}{R_{th}} = \frac{2\pi}{l}\left[\frac{K_1 x^3}{3} + \frac{K_2 x^2}{2}\right]_0^R$$

$$\frac{1}{R_{th}} = \frac{2\pi}{l}\left(\frac{K_1 R^3}{3} + \frac{K_2 R^2}{2}\right)$$

$$\frac{1}{R_{th}} = \frac{2\pi}{l}\left(\frac{2K_1 R^3 + 3K_2 R^2}{6}\right)$$

$$\boxed{R_{th} = \frac{3l}{\pi(2K_1 R^3 + 3K_2 R^2)}}$$

* Rate of heat flow,

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$$P = \frac{\Delta T}{R_{th}} = \frac{\pi (2K_1 R^3 + 3K_2 R^2)}{3l} (\Delta T)$$

* Effective thermal conductivity,

M-I: $R_{th} = \frac{l}{K_{eff} A}$

$$\frac{3l}{\pi R^2 (2K_1 R + 3K_2)} = \frac{l}{K_{eff} A}$$

$$\boxed{K_{eff} = \frac{2K_1 R + 3K_2}{3}}$$

M-II:

$$\frac{K_p A_1}{l} + \frac{K_p A_2}{l_2} + \cdots = \frac{K_1 A_1}{l} + \frac{K_2 A_2}{dx} + \cdots$$

$$K_p = \frac{\sum K_i A_i}{\sum A_i} = \frac{\int K \cdot dA}{\int dA}$$

$$K_p = \frac{\int_0^{x=R} (K_1 x + K_2) 2\pi x (dx)}{\pi R^2}$$

$$= \frac{2\pi \left[K_1 \frac{R^3}{3} + K_2 \frac{R^2}{2} \right]}{\pi R^2}$$

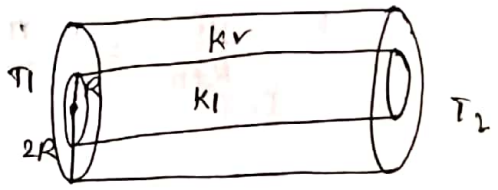
$$= \frac{2K_1 R}{3} + K_2$$

$$\boxed{K_p = \frac{2K_1 R + 3K_2}{3}}$$

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Ex:

Find effective K-value



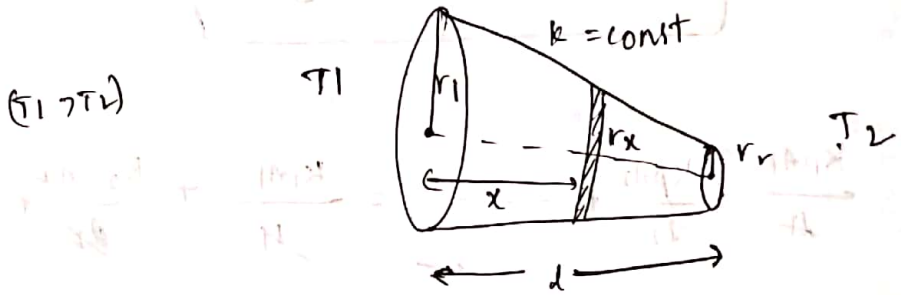
Sol:

$$K_{eff} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$$

$$= \frac{K_1(\pi R^2) + K_2(4\pi R^2 - \pi R^2)}{5\pi R^2 - \pi R^2}$$

$$\boxed{K_{eff} = \frac{K_1 + 3K_2}{4}}$$

Ex:



case①: Radius of rod is decreasing linearly from r_1 to r_2. Then,

$$\boxed{r_x = r_1 - \left(\frac{r_1 - r_2}{l}\right)x}$$

∴ Thermal resistance of rod,

$$R_{th} = \int dR$$

$$R_{th} = \int \frac{dx}{kA}$$

$$= \int \frac{dx}{k\pi\left(r_1 - \left(\frac{r_1 - r_2}{l}\right)x\right)^2}$$

on simplifying,

$$\boxed{R_{th} = \frac{l}{k\pi r_1 r_2}}$$

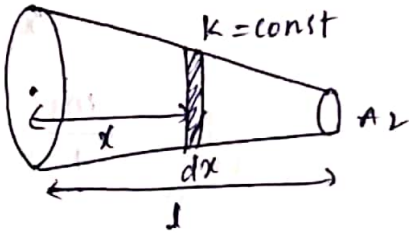
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* Rate of heat flow,

$$P = \frac{\Delta T}{R_{th}} = \frac{(\Delta T) K \pi r_1 r_2}{l}$$

case ②: If area of cross-section is decreasing linearly from A_1 to A_2.

$$A_2 = A_1 - \left(\frac{A_1 - A_2}{l}\right) x$$



Effective thermal resistance,

$$R_{th} = \int \frac{dx}{KA}$$

$$= \int_{x=0}^{x=R} \frac{dx}{K\left[A_1 - \left(\frac{A_1 - A_2}{l}\right)x\right]}$$

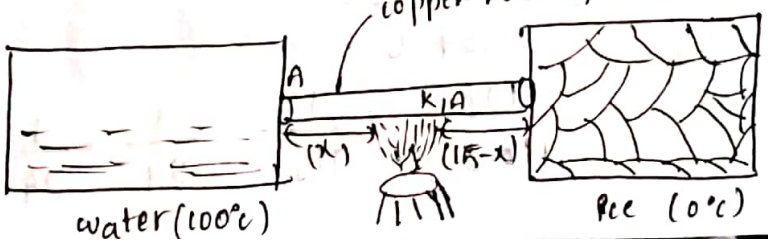
$$= \frac{l}{k(A_1 - A_2)} \left\{ \ln \left[A_1 - \frac{A_2}{l} \right] x \right\}_{x=0}^{x=l}$$

$$R_{th} = \frac{l}{k(A_1 - A_2)} \ln (A_1/A_2)$$

* rate of heat flow,

$$P = \frac{\Delta T}{R_{th}} = \frac{K(A_1 - A_2)\Delta T}{l \ln(A_1/A_2)}$$

Ex: copper rod of length 1.5m



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where should a temp of 200°c is maintained in the rod so that, mass of ice melts is equal to mass of water that converts into steam in the same time. Neglect heat loss to the surroundings.

Soll

$$Q_1 = \frac{KA(200-100)t}{x} = m_s L_s$$

$$\frac{m_s}{t} = \frac{KA(100)}{x(L_s)} \quad -①$$

and

$$Q_2 = \frac{KA(200-0)t}{15-x} = m_i L_i$$

$$\frac{m_i}{t} = \frac{KA \times 200}{(1.5-x)L_i} \quad -②$$

Given, $\frac{m_i}{t} = \frac{m_s}{t}$

$$\frac{200KA}{(1.5-x)L_i} = \frac{100KA}{xL_s}$$

$$2xL_s = 1.5L_i - xL_i$$

$$(2L_s + L_i)x = 1.5L_i$$

$$x = \frac{1.5L_i}{2L_s + L_i}$$

$$= \frac{1.5 \times 80}{1080 + 80}$$

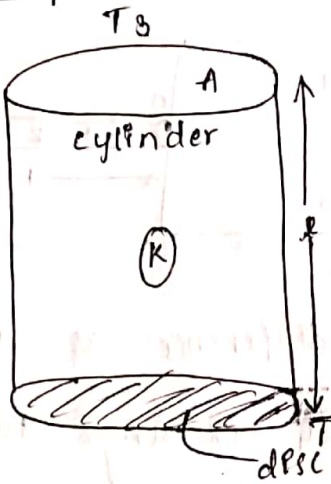
$$= \frac{120}{1160} = \frac{12}{116} = \frac{3}{29} = \frac{3}{29} m$$

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$$x = \frac{300}{29}\ cm$$

$$\boxed{x = 10.34 cm}$$

Ex: cylinder is kept co-axially ~~constantly~~ on a disc. Initially disc is at temp. T_1 ($T_3 > T_1$). Find the time taken by the disc to increase its temp from T_1 to T_2.



Sol: clearly, $\boxed{T_3 > T_2 > T_1}$

at any intermediate state, T is temp of disc.

In next dt sec,

$$\frac{dQ}{dt} = \frac{KA(T_3 - T)}{l}$$

heat absorbed by the disc, $dQ = ms\,dT$

$$\therefore ms\frac{dT}{dt} = \frac{KA(T_3 - T)}{l}$$

$$\int_{T_1}^{T_2} \frac{dT}{T_3 - T} = \int_0^t \frac{KA}{msl}\cdot dt$$

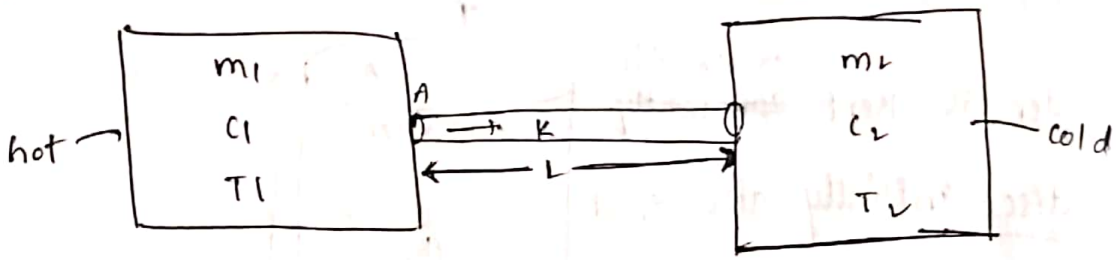
$$-\ln(T_3 - T_2 / T_3 - T_1) = \frac{KA}{msl}t$$

$$\ln\left(\frac{T_3 - T_1}{T_3 - T_2}\right) = \frac{KAt}{msl}$$

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$$t = \frac{msl}{KA} \ln\left(\frac{T_3 - T_1}{T_3 - T_2}\right)$$

Ex:



Find temp difference as a function of time 't', neglecting heat exchange to it's surrondings.

Soln

At any intermediate state,

$$m_1 c_1 \frac{dT_1}{dt} = -\frac{KA(T_1 - T_2)}{L} \quad (\Delta T)$$

$$\frac{dT_1}{dt} = -\frac{KA}{m_1 c_1 L}(T_1 - T_2) \quad (\Delta T) \quad -(1)$$

and $$m_2 c_2 \frac{dT_2}{dt} = -\frac{KA(T_1 - T_2)}{L} \quad (\Delta T)$$

$$\frac{dT_2}{dt} = \frac{-KA(T_1 - T_2)}{m_2 c_2 L} \quad (\Delta T) \quad -(2)$$

eqn (1) - (2),

$$\frac{d}{dt}(T_1 - T_2) = -\frac{KA}{L}(T_1 - T_2)\left(\frac{1}{m_1 c_1} + \frac{1}{m_2 c_2}\right) \quad (\Delta T)$$

$$\int_{(T_1-T_2)_i}^{(T_1-T_2)_f} \frac{d(T_1 - T_2)}{(T_1 - T_2)} = -\frac{KA}{L}\left(\frac{1}{m_1 c_1} + \frac{1}{m_2 c_2}\right)\int_0^t dt \quad (\Delta T)$$

ΔT_i $(T_1 - T_2)_i$

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$$\left[\ln (T_1 - T_2)\right]_{\Delta T_i}^{\Delta T_f} = \frac{-KA}{L}\left(\frac{1}{m_1 c_1} + \frac{1}{m_2 c_2}\right) t$$

$$\ln\left(\frac{(T_1 - T_2)_f}{(T_1 - T_2)_i}\right) = \frac{-KA}{L}\left(\frac{1}{m_1 c_1} + \frac{1}{m_2 c_2}\right) t$$

Correction:

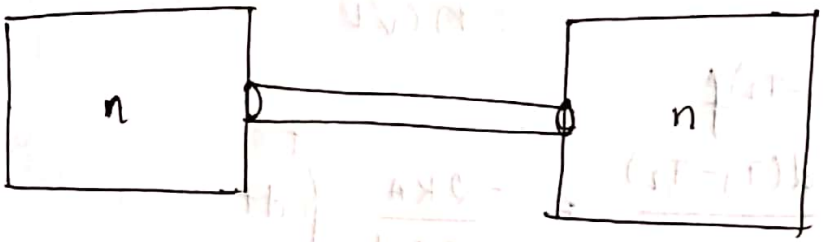
$$\Delta T_f = \Delta T_i \cdot e^{\frac{-KAt}{L}\left(\frac{1}{m_1 c_1} + \frac{1}{m_2 c_2}\right)}$$

Correct answer

$$(T_1 - T_2)_f = (T_1 - T)_i \cdot e^{\frac{-KAt}{L}\left(\frac{1}{m_1 c_1} + \frac{1}{m_2 c_2}\right)}$$

ΔT ΔT

Ex!



n-moles of ideal diatomic gas is taken in both the closed vessels. Find the temp diff as a function of time?

Sol!! we have,

$$dQ = dU + \cancel{dW}^{0}$$

$$\boxed{dQ = n C_v dT} \quad \text{where} \quad \boxed{C_v = \frac{5R}{2}}$$

at any intermediate state,

$$ncv \frac{dT_1}{dt} = \frac{-KA(T_1 - T_2)}{L}$$

$$\frac{dT_1}{dt} = \frac{-KA(T_1 - T_2)}{ncv L} \quad -(1)$$

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and $nC_V \frac{dT_2}{dt} = +\frac{KA(T_1-T_2)}{L}$

$$\frac{dT_2}{dt} = +\frac{KA(T_1-T_2)}{nC_V L} \quad -②$$

eqn ① - ②

$$\frac{d}{dt}(T_1-T_2) = -\frac{2KA(T_1-T_2)}{nC_V L}$$

$$\int_{(T_1-T_2)_i}^{(T_1-T_2)_f} \frac{d(T_1-T_2)}{T_1-T_2} = -\frac{2KA}{nC_V L}\int_0^t dt$$

$$\ln\left[(T_1-T_2)_f/(T_1-T_2)_i\right] = \frac{-2KA}{nC_V L}t$$

$$(T_1-T_2)_f = (T_1-T_2)_i\, e^{\frac{-2KAt}{n(5R/2)L}}$$

Correction!

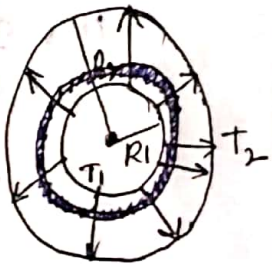
$\Delta T_f = \Delta T_i \cdot e^{\frac{-4KAt}{5nR}}$

$$(T_1-T_2)_f = (T_1-T_2)_i\, e^{\frac{-4KAt}{5nR}}$$

Radial flow of heat:

① Concentric spherical shells:

Gap blw the shells is filled with a material of thermal conductivity.



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here $T_1 > T_2$.

In steady state,

$$P = -KA \frac{dT}{dr}$$

$$P = -K 4\pi r^2 \cdot \frac{dT}{dr}$$

$$\int_{R_1}^{R_2} \frac{dr}{r^2} = -\int_{T_1}^{T_2} \frac{4\pi K}{P} dT$$

$$\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{4\pi K}{P} (T_1 - T_2)$$

Rate of heat flow

$$\boxed{P = \frac{4\pi K (T_1 - T_2) R_1 R_2}{R_2 - R_1}}$$

→ rate of heat flow

Clearly $P \propto \frac{R_1 R_2}{R_2 - R_1}$

* Thermal resistance (R_{th}):

we have, $\frac{\Delta T}{R_{th}} = P = \frac{T_1 - T_2}{\frac{R_2 - R_1}{4\pi K R_1 R_2}}$

$$\boxed{R_{th} = \frac{R_2 - R_1}{4\pi K R_1 R_2}}$$

$$\Rightarrow R_{th} \propto \frac{R_2 - R_1}{R_1 R_2}$$

* The pt b/w the shells where the temp is half way of the sum of temps of inner & outer shells. ie,

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$$T = \frac{T_1 + T_2}{2}$$

$$\therefore P = \frac{T_1 - T}{R_1} = \frac{T - T_2}{R_2}$$

$$\Rightarrow \boxed{R_1 = R}$$

~~$\frac{r - R_1}{4\pi K r R_1} = \frac{R_2 - r}{4\pi K r R_2}$~~

$$\frac{r - R_1}{4\pi K r R_1} = \frac{R_2 - r}{4\pi K r R_2}$$

$$R_2(r - R_1) = (R_2 - r)R_1$$

$$R_2 r - R_1 R_2 = R_2 R_1 - r R_1$$

$$r(R_2 + R_1) = R_1 R_2$$

$$\boxed{r = \frac{R_1 R_2}{R_1 + R_2}}$$

Temp distribution:

We have,

$$P = -KA\frac{dT}{dr}$$

$$P = -4\pi K r^2 \cdot \frac{dT}{dr}$$

$$\int_{R_1}^{R_2} \frac{dr}{r^2} = -\frac{4\pi K}{P}\int_{T_1}^{T_2} dT$$

$$\left[-\frac{1}{r}\right]_{R_1}^{R_2} = -\frac{4\pi K}{P}\left[T_2 - T_1\right]$$

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{4\pi K}{P}(T_1 - T_2)$$

$R_2 = r, \quad T_2 = T$

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$$T_1 - T = \frac{P}{4\pi K}\left[\frac{r-R_1}{rR_1}\right]$$

$$\boxed{T = T_1 - \frac{P}{4\pi K}\left(\frac{r-R_1}{rR_1}\right)}$$

where $P = \dfrac{4\pi K R_1 R_2 (T_1 - T_2)}{R_2 - R_1}$

$$T = T_1 - \frac{R_2 R_1 (T_1-T_2)}{R_2-R_1}\left(\frac{r-R_1}{rR_1}\right)$$

$$T = T_1 - \frac{R_2(r-R_1)(T_1-T_2)}{r(R_2-R_1)}$$

$$= \frac{rR_2T_1 - rR_1T_1 - (R_2r - R_1R_2)(T_1-T_2)}{r(R_2-R_1)}$$

$$T = \frac{rR_2T_1 - rR_1T_1 - R_2rT_1 - R_1R_2T - R_2rT_2 + R_1R_2T_2}{r(R_2-R_1)}$$

$$rR_2T - rR_1T = R_1R_2T_2 - rR_2T_2 - rR_1T_1 - R_1R_2T$$

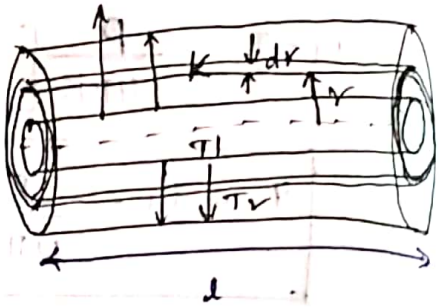
$$T(rR_2 - rR_1 + R_1R_2) = (R_1R_2 + rR_2)T_2 - rR_1T_1$$

$$\boxed{T = \frac{(R_1R_2 - rR_2)T_2 - rR_1T_1}{(rR_2 - rR_1 + R_1R_2)}}$$

finally, $T =$

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(2) Concentric Cylindrical shells:



Gap b/w the shells is filled with a material of thermal Conductivity K.

Here $T_1 > T_2$

In steady state,

$$P = -KA\frac{dT}{dr}$$

$$P = -K(2\pi rl)\frac{dT}{dr}$$

$$-\frac{P}{2\pi Kl} = r\frac{dT}{dr}$$

$$\int_{R_1}^{R_2}\frac{dr}{r} = -\frac{2\pi Kl}{P}\int_{T_1}^{T_2}dT$$

$$\ln(R_2/R_1) = -\frac{2\pi Kl}{P}(T_2-T_1)$$

$$\ln(R_2/R_1) = \frac{2\pi Kl}{P}(T_1-T_2)$$

Rate of heat flow,

$$\boxed{P = \frac{2\pi Kl(T_1-T_2)}{\ln(R_2/R_1)}}$$

* Thermal resistance (R_{th}):

we have $P = \frac{\Delta T}{R_{th}} = \frac{T_1-T_2}{\frac{\ln(R_2/R_1)}{2\pi Kl}}$

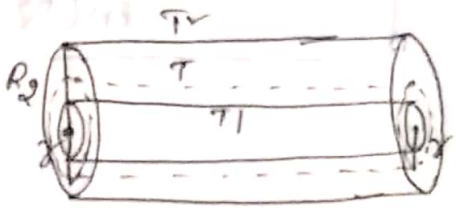
$$\Rightarrow \boxed{R_{th} = \frac{\ln(R_2/R_1)}{2\pi Kl}}$$

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* The pt blw the shells where $T = \frac{T_1 + T_2}{2}$ is

In series combination:

In steady state,



$$P = \frac{T_1 - T}{R_1} = \frac{T - T_2}{R_2}$$

$$R_1 = R_2$$

$$\Rightarrow \frac{\ln(r/R_1)}{2\pi K l} = \frac{\ln(R_2/r)}{2\pi K l}$$

$$\frac{r}{R_1} = \frac{R_2}{r}$$

$$r^2 = R_1 R_2$$

$$\boxed{r = \sqrt{R_1 R_2}}$$

Temp distribution:

we have,

$$\int_{R_1}^{r} \frac{dr}{r} = \frac{-2\pi K l}{P} \int_{T_1}^{T} dT$$

$$\ln(r/R_1) = -\frac{2\pi K l}{P}(T - T_1)$$

$$\ln(r/R_1) = \frac{2\pi K l}{P}(T_1 - T)$$

~~$\neq$~~

$$T_1 - T = \frac{P \ln(r/R_1)}{2\pi K l}$$

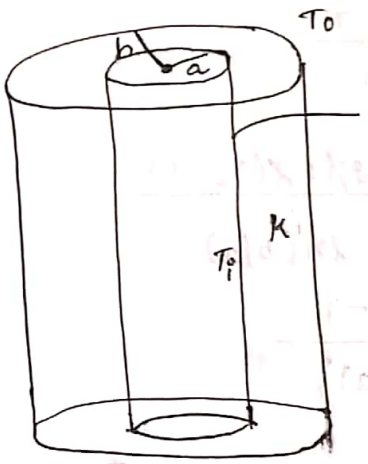
$$T = T_1 - \frac{\ln(r/R_1) P}{2\pi K l}$$

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$$T = T_1 - \frac{\ln(r/R_1)}{\ln(R_2/R_1)} (T_1 - T_2)$$

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Ex!



inner solid cylinder is having mass 'm' & heat capacity per unit volume is "S".

Its initial temp is T_1 ($< T_2$).

Find the time required by inner cylinder to increase its temp from T_1 to T_2.

Sol! clearly $T_0 > T_2 > T_1$

At any intermediate state, temp of inner cylinder = T.

In steady state,

$$P = KA \frac{dT}{dx}$$

$$P = K(2\pi r l) \cdot \frac{dT}{dr}$$

$$\int_b^a \frac{dr}{r} = \frac{2\pi K l}{P} \int_{T_0}^{T} dT$$

$$\boxed{P = \frac{2\pi K l (T_0 - T)}{\ln(b/a)}}$$

$$\left(\frac{dQ}{dt}\right)_{time} = \frac{2\pi K l (T_0 - T)}{\ln(b/a)}$$

we have heat absorbed by inner cylinder.

$$dQ = m s_{sp} dT$$

Given $$\left(\frac{dQ}{dT}\right)_{temp} = \pi a^2 l (s) = m s_{sp}$$

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$$m s_{sp} \frac{dT}{dt} = \frac{2\pi K l (T_0 - T)}{\ln(b/a)}$$

$$\pi a^2 l (s) \left(\frac{dT}{dt}\right) = \frac{2\pi K l (T_0 - T)}{\ln(b/a)}$$

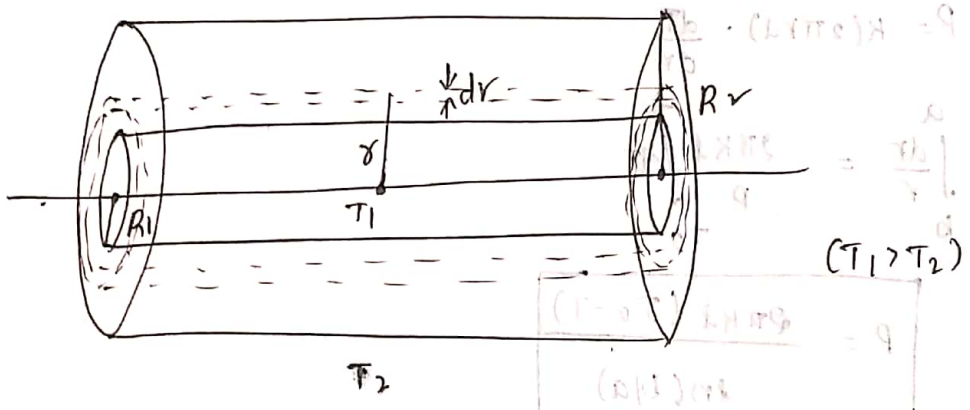
$$\frac{dT}{dt} = \frac{2K(T_0 - T)}{a^2 (\ln(b/a))}$$

$$\int_{T_1}^{T_2} \frac{dT}{T_0 - T} = \frac{2K}{a^2 s \ln(b/a)} \int_0^t dt$$

$$\ln\left(\frac{T_0 - T_1}{T_0 - T_2}\right) = \frac{2K}{a^2 s \ln(b/a)} \times t$$

$$\boxed{t = \frac{a^2 s \ln(b/a) \ln\left(\frac{T_0 - T_1}{T_0 - T_2}\right)}{2K}}$$

13/11/24 Wednesday



Thermal conductivity of the material that fills the gap b/w the shells varies with distance r from axis is $\boxed{K = \frac{\alpha}{r^2}}$

In steady state,

$$P = -KA \frac{dT}{dr}$$

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$$= -k 2\pi r l \cdot \frac{dT}{dr}$$

$$\Rightarrow P = -\frac{\alpha}{r^2} \times 2\pi r l \frac{dT}{dr}$$

$$\int_{R_1}^{R_2} r \cdot dr = -\int_{T_1}^{T_2} \frac{2\pi l \alpha}{P} \cdot dT$$

$$\frac{R_2^2 - R_1^2}{2} = \frac{2\pi l \alpha}{P} (T_1 - T_2)$$

$$\boxed{P = \frac{4\pi \alpha l (T_1 - T_2)}{R_2^2 - R_1^2}} \quad \text{--- (a)}$$

Thermal ~~conductivity~~ Resistance (R_{th}) :-

M-I : $R_{th} = \frac{\Delta T}{P}$

$$\boxed{R_{th} = \frac{R_2^2 - R_1^2}{4\pi \alpha l}}$$

M-II : $R_{th} = \int dR$ (since elementary shells are in series)

$$= \int \frac{dr}{KA} = \int \frac{dr}{\frac{\alpha}{r^2}(2\pi r l)}$$

$$\boxed{R_{th} = \frac{R_2^2 - R_1^2}{4\pi \alpha l}}$$

Temp distribution :-

we have,

$$P \int_{R_1}^{r} r \cdot dr = -2\pi \alpha l \int_{T_1}^{T} dT$$

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$$P\left(\frac{r^2-R_1^2}{2}\right) = 2\pi r l (T_1 - T)$$

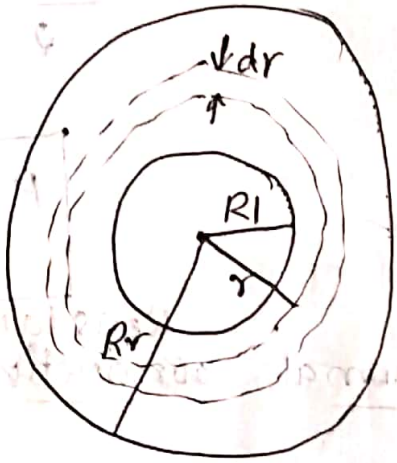
$$T_1 - T = \frac{P}{4\pi r l}(r^2 - R_1^2)$$

$$\boxed{T = T_1 - \frac{(T_1 - T_2)(r^2 - R_1^2)}{R_2^2 - R_1^2}} \quad \text{(from eq@)}$$

thermal conductivity,

$$\boxed{K = \frac{\alpha}{r^2}}$$

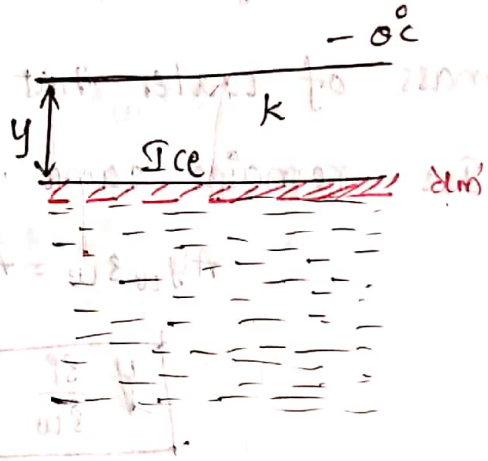
$$\boxed{K = \frac{2\alpha l \ln(R_2/R_1)}{R_2^2 - R_1^2}}$$



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Formation of Ice in Lakes and ponds:

At any instant 't', y is the thickness of Ice (taken from surface) in next dt sec.



$$\frac{dQ}{dt} = \frac{KA(0-(-\theta))}{y}$$

$$dQ = dm L_I = A\,dy\,\rho_w L_I$$

$$\frac{A\,dy\,\rho_w L_I}{dt} = \frac{KA\theta}{y} \Rightarrow \int_0^y y\,dy = \frac{K\theta}{\rho_w L_I}\int_0^t dt$$

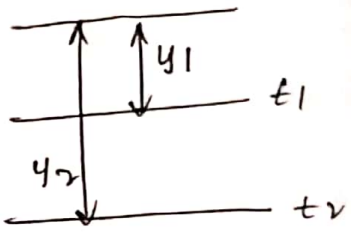
$$\frac{y^2}{2} = \frac{K\theta}{\rho_w L_I} t$$

$$\Rightarrow \boxed{t = \frac{y^2 \rho_w L_I}{2K\theta}} \Rightarrow (t \propto y^2)$$

+ve

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Here $t_1 = \frac{y_1^2 \rho_i L_I}{2K\theta}$



$$t_2 = \frac{y_2^2 \rho_i L_I}{2K\theta}$$

time taken by the ice to increase its thickness from y_1 to y_2 —

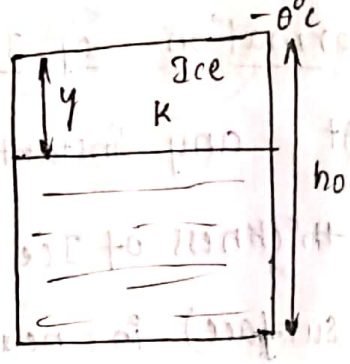
$$\boxed{\Delta t = \frac{(y_2^2 - y_1^2)\rho_i L_I}{2K\theta}}$$

time taken ratios,

$$\Delta t_1 : \Delta t_2 : \Delta t_3 = (y_1^2 - y_1^2) : (y_3^2 - y_2^2) : (y_4^2 - y_3^2) \ldots$$

initial height of water $= h_0$

mass of water that converts into Ice remains same



$$A y_w \rho_w = A y \rho_I$$

$(\rho_i < \rho_w)$

$$\boxed{y \frac{\rho_i}{\rho_w} = y_w}$$

$(y_w < y)$

increase in height of column

$$\Delta y = y - y_w$$

$$= y - y\frac{\rho_i}{\rho_w}$$

$$= y\left(1 - \frac{\rho_i}{\rho_w}\right)$$

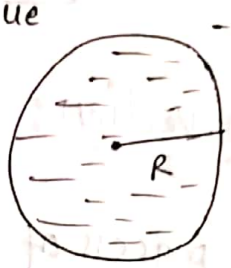
$$= \sqrt{\frac{2K\theta t}{\rho_w L_i}}\left(1 - \frac{\rho_i}{\rho_w}\right)$$

$\therefore$ height of the column at time t,

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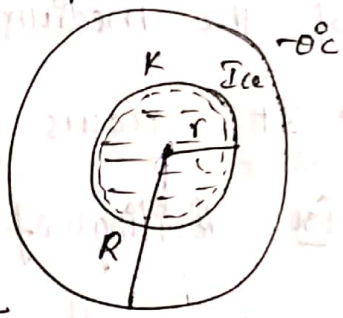
$$h = h_0 + \Delta y = h_0 + \sqrt{\frac{2K\theta t}{\rho_w L_I}\left(1 - \frac{\rho_i}{\rho_w}\right)}$$

Ex: water fills the spherical shell due to $-\theta^\circ C$ temp of entire water cools to $0^\circ C$ and then starts freezing from outer surface.



Highly conducting spherical shell

At any instant 't'



$$R_{th} = \frac{R-r}{4\pi K r R}$$

In the next dt sec

$$\frac{d\theta}{dt} = \frac{\Delta T}{R_{th}} = \frac{(0-(-\theta))\,4\pi K r R}{(R-r)}$$

$$dQ = dm L_I = 4\pi r^2 dr\, \rho_w L_I$$

$$-\frac{4\pi r^2 dr\, \rho_w L_I}{dt} = \frac{4\pi K r R \theta}{(R-r)}$$

$$\Rightarrow -\int r(R-r)\,dr = \frac{KR\theta}{\rho_w L_I}\int dt$$

-ve indicates 'r' decreases with time.

$$-\int_R^0 r(R-r)\cdot dr = \frac{KR\theta}{\rho_w L_I}\int_0^t dt$$

$$\frac{R^3}{2} - \frac{R^3}{3} = \frac{KR\theta}{\rho_w L_I}\, t$$

$$\frac{R^2}{6} = \frac{Kt}{\rho_w L_I}$$

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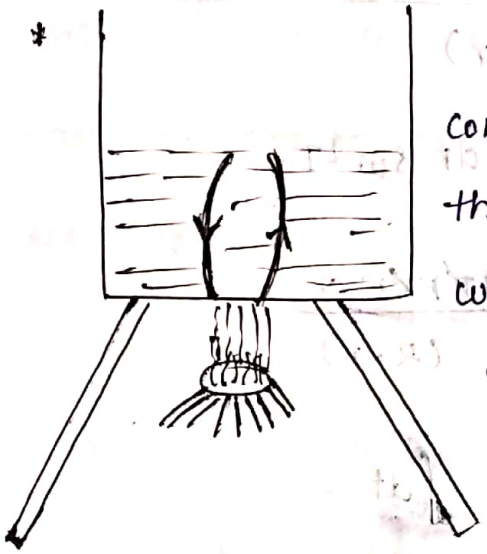
$$t = \frac{R^4 \rho \omega L_I}{6K\theta}$$

Convection:

The process of heat transfer from one place to another place by the actual movement of particles of the medium.

* It occurs in fluids.

Ex: Boiling of water.



* water molecules absorb heat from container & become less densar. so they move to top & the molecules which are densar move to bottom & again absorb heat.

Types of convection:

① Natural Convection:

It occurs due to difference of density on heating. in this process "gravity" plays an important role.

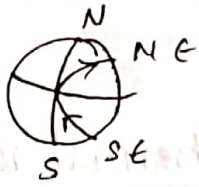
Ex: ① Boiling of water

(2) Sea Breeze & Land breeze.

(3) Trade winds

(4) Monsoons

(5) ventilaters



(2) forced Convection:

In Natural convection rate of flow of heat is directly proportional to ~~Area~~

$$\frac{d\theta}{dt} \propto A$$ (Area of surface that exposed to fluid)

$$\frac{d\theta}{dt} = \Delta\theta$$ (temp diff b/w surface & fluid)

$$\boxed{\frac{d\theta}{dt} = hA\Delta\theta}$$

h = coefficient of convection.

Ex:

Radiation:

The process of transmission of heat from one place to another without the necessicity of the material medium.

In this process heat transmittes in the form of "electromagnetic radiations"

Ex: Heat transmittes from sun to earth.

Thermal radiation:

The emission of electromagnetic waves at the expense of the internal energy (which the function

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of temp only) of the bodies is known as thermal radiation.

Properties of thermal radiation:

(1) It travels along a st. lines like light with speed = 3×10^8 in air ~~medium~~ medium.

(2) It can travel even in vaccum. wave length of radiation is. Ranging from $7800A^\circ$ – 1mm (infrared radiation.

Thermal radiation obeys inverse square law i.e intensity of heat at a pt varies inversly as the square of the distance of that pt from the source.

$$I \propto \frac{1}{r^2}$$

$$\boxed{\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}}$$

radiation exert a small but finite pressure on the surface on which it falls this is called radiation pressure.

Applications of Thermal radiations:

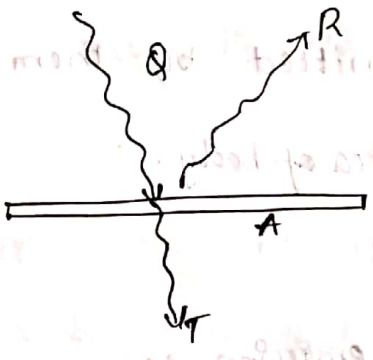
1) A thermogram is used to locate the broken part of a bone. Broken part of bone radiates more heat than other parts.

2) Infra red detectors are used as night scopes and also detect radiation from skin.

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3) Thermography is used to screen travellers at air ports who suffer from high fever carrying infection with them.

Reflectance, absorbtance, transmittence:



$$Q = R + T + A$$

$$\frac{R}{Q} + \frac{T}{Q} + \frac{A}{Q} = 1$$

$$r + t + a = 1$$

for a specific wavelength $\boxed{r_\lambda + t_\lambda + a_\lambda = 1}$

$r = \frac{R}{Q}$ = reflectance (or) reflection coefficient

$a = \frac{A}{Q}$ = absorbtion coeff

$t = \frac{T}{Q}$ = transmission coeff

} No units

points:

① If $t = 0$ (no transmitted energy),

$\boxed{a + r = 1} \Rightarrow$ If $a = 1 \Rightarrow r = 0$

i.e, poor reflector (or) good absorber of heat.

Ex: Black Body.

② If $a = 0$, $r = 0$, $\boxed{t = 1}$

③ If $t = 0$, $a = 0$, $\boxed{r = 1}$

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Emissive Power :

All bodies emitte heat radiations at all temp. the energy emitted by them consists of all wave lengths ranging from 0 to ∞.

Amount of energy emitted by them depends on

1) The nature of and area of body.
2) Temp of body
3) the time for which emission occurs.

The same body radiates energy differently at different temp.

Spectral Emissive power (E_λ) : (for a particular wavelength)

E_λ of a body at a temp 'T' for a wavelength 'λ' is defined as the energy radiated in vaccum per unit time per unit area & per unit range of wave lengths (lying b/w λ & λ+dλ)

$$E_\lambda = \frac{\text{Energy}}{\text{Area} \times \text{time} \times \text{wavelength}}$$

$\frac{J}{m^2 \text{ sec } A^\circ}$

E_λ for body is different for diff values of λ & also for diff values of T.

Total emissive Power :

It is defined as the total amount of Energy

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radiated per unit time per unit area of body for all possible wavelengths at a given temp.

$$\epsilon = \frac{\text{Energy}}{\text{Area} \times \text{time}}$$

* Emissive power is max for Black body

Emissivity: (e)- 14|11|24|Thursday

At a given temp the ratio of spectral emissive power of a body to that of spectral emissive power of black body.

$$e = \frac{\epsilon_\lambda}{\epsilon_{\lambda b}} \quad (\text{or}) \quad e = \frac{\epsilon}{\epsilon_b}$$

* e has no units & dimensions.

$0 < e < 1$.

* For black body, $e = 1$

~~(Absorbtive Power:)~~

Energy Flux:

energy radiated per second is called energy flux.

$$\text{Energy flux} = \frac{\text{Energy}}{\text{time}}$$

**Emissive power $= \frac{\text{Energy flux}}{\text{Area}}$

Absorbtive Power:

$$A_\lambda = \frac{\text{Energy flux absorbed per unit area of unit wavelength range.}}{\text{energy flux striking per unit area of same wavelength range.}}$$

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$$A_\lambda = \frac{dQ_{x'}}{dQ_x}$$

Total absorbtive power,

$$A = \int_0^\infty A_\lambda \, d\lambda$$

* for black body, $A = 1$

Kirchoff's Law:

$$\frac{E_\lambda}{A_\lambda} = \frac{E_{\lambda b}}{A_{\lambda b}} = \text{const}$$

* The ratio of emissive power to absorbtive power is same for all bodies at particular temp.

Since $E_{\lambda b}$ increases with temp,

$$\Rightarrow \frac{\epsilon_\lambda}{A_\lambda} \ (\uparrow)$$

$$\Rightarrow A_\lambda = \frac{\epsilon_\lambda}{\epsilon_{\lambda b}} = e_\lambda$$

* $\epsilon_\lambda \propto A_\lambda$

good emitters are good absorbers

Prevost's Theory of Heat Exchanges:

① All the bodies radiate heat to the surrounding and absorbs heat radiation from the surrounding simultaneously at all temperatures except at absolute 0 K.

② If temp of body is more than surrounding temp then it absorb less heat & radiates more heat.

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③ If temp of body is less than surrounding temp then it absorbs more heat & radiates less heat.

* radiation of heat from a body depends on its temp but absorbtion of heat to body depends on the surrounding temp.

④ Temp of body = surrounding temp, the absorbtion & radiation is same.

Stefan's Law :

Emissive power of a black body is directly proportional to "fourth power of it's absolute temp".

$$E \propto T^4$$

$$E = \sigma T^4$$

where, σ = stefen's const

$$= 5.67 \times 10^{-8} \ \frac{W}{m^2 K^4}$$

$$\frac{Q}{At} = \sigma T^4$$

Radiating power is,

$$P = \frac{Q}{t} = \sigma A T^4 \quad **$$

∴ energy radiated by body in t-secs,

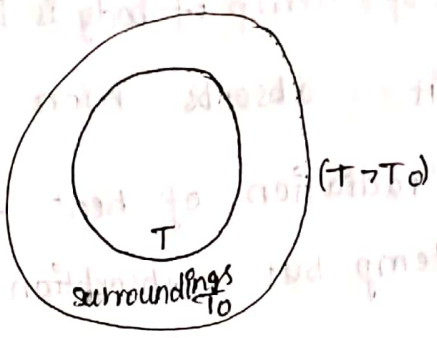
$$Q = \sigma A T^4 t$$

* for other than black body,

$$P = \sigma e A T^4$$

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Stefan - Boltzmann's Law:



* Rate of heat radiated by the body to the surrounding

$$P_1 = \sigma e A T^4$$

* Rate of heat absorbed by the body,

$$P_2 = \sigma e A T_0^4$$

* net rate of loss of heat of the body,

$$P = P_1 - P_2$$

$$\therefore \boxed{\frac{dQ}{dt} = \sigma e A (T^4 - T_0^4)} \quad ***$$

* If $T < T_0$ then net rate of gain of heat of body,

$$\boxed{\frac{dQ}{dt} = \sigma e A (T_0^4 - T^4)}$$

we have,

$$dQ = ms\,dT$$

$$-ms\frac{dT}{dt} = \sigma e A (T^4 - T_0^4)$$

$\therefore$ rate of fall in temp / rate of cooling of body,

$$\boxed{\frac{-dT}{dt} = \frac{\sigma e A}{ms} (T^4 - T_0^4)}$$

here mass = vol x density

Eg:
for sphere - $A = 4\pi R^2$

$V = \frac{4}{3}\pi R^3$

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$$\frac{A}{m} = \frac{A}{V\rho} = \frac{4\pi r^2}{\frac{4}{3}\pi r^3 \times \rho} = \frac{3}{r\rho}$$

Ex: consider a sphere, cube & a thin circular plate of same mass & material, heated to same temp. which one of these will have the fastest rate of cooling.

Sol: we have,

$$\frac{-dT}{dt} = \frac{\sigma e A}{ms}(T^4 - T_0^4)$$

here $m, s, e, T, T_0 \longrightarrow$ same

$$\Rightarrow \boxed{-\frac{dT}{dt} \propto A}$$

for a given volume,

surface area is minimum for sphere.

Surface area is maximum for disc.

Ex: A solid copper cube & a sphere both of same mass & emissivity are heated to same initial temp and kept under identical condns. what is the ratio of their inital rate of the fall of temp.

Sol:

$$-\frac{dT}{dt} = \frac{\sigma e A}{ms}(T^4 - T_0^4)$$

$$\left(-\frac{dT}{dt}\right) \propto A$$

$$\frac{\left(-\frac{dT}{dt}\right)_{cube}}{\left(-\frac{dT}{dt}\right)_{sphere}} = \frac{A_{cube}}{A_{sphere}} = \frac{6L^2}{4\pi r^2} = \left(\frac{6}{\pi}\right)^{1/3}$$

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$$\frac{1.4}{\left(\frac{dT}{dt}\right)_2} = \frac{\sigma(T_1^4 - T_0^4)}{T_2^4 - T_0^4}$$

$$\frac{1.4\,(T_2^4 - T_0^4)}{T_1^4 - T_0^4} = \left(\frac{dT}{dt}\right)_2$$

$$\frac{1.4\,(6^4 - 3^4)}{4^4 - 3^4} = \left(\frac{-dT}{dt}\right)_2$$

$$\frac{1.4(1215)}{175} = \left(\frac{dT}{dt}\right)_2 \Rightarrow \boxed{\left(\frac{dT}{dt}\right)_2 = 9.72\,^{\circ}C/min}$$

Ex: Consider a block of copper of radius 5cm (spher... Its outer surface is coated black. Howmuch time is required for the sphere to cool down from 1000K to 300K. $\rho_{cu} = 9 \times 10^3$ Kg/m^3 & Specific heat 4 KJ/KgK.

Sol^n

we have,

$$\frac{dQ}{dt} = \sigma \epsilon A T^4 \qquad (\epsilon = 1)$$

$$-ms\frac{dT}{dt} = \sigma A T^4$$

$$-\int_{1000}^{300} \frac{dT}{T^4} = \frac{\sigma}{s} \times \frac{3}{r\rho} \int_0^t dt$$

$$-\left[-T^{-3}\right]_{1000}^{300} = \frac{3\sigma}{rs\rho}\, t$$

$$\left[\frac{1}{T^3}\right]_{1000}^{300} = \frac{3\sigma}{rs\rho}\, t$$

$$\left[\frac{1}{27\times10^6} - \frac{1}{10^9}\right] = \frac{8 \times 5.67\times10^{-8}}{5\times10^{-2} \times 4\times10^3 \times 9\times10^3}\, t$$

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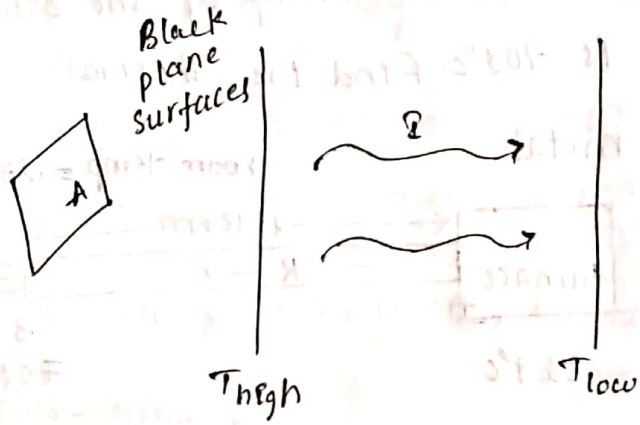
$$\frac{1000 - 27}{9 \times 10^{-2}} = \frac{5.67 \times 10^{-6}}{2} \times t$$

$$t = \frac{973 \times 2}{9 \times 5.67 \times 10^{-4}}$$

$$= \frac{973 \times 2}{9 \times 5.67} \times 10^{4}$$

$t =$

*** **Heat shielding problem :**



net rate of flow of heat,

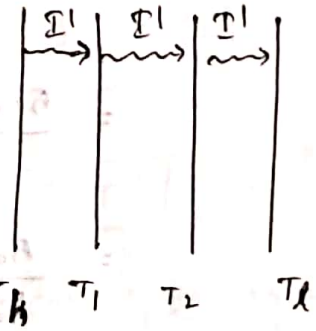
$$I = \sigma A (T_h^4 - T_l^4) \quad \text{—①}$$

* After steady state condn achieved,

$$I' = \sigma A (T_h^4 - T_1^4)$$

$$I' = \sigma A (T_1^4 - T_2^4)$$

$$I' = \sigma A (T_2^4 - T_l^4)$$



on adding,

$$3I' = \sigma A (T_h^4 - T_l^4)$$

$$\boxed{I' = I/3}$$

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***

Heat absorbed (rate) = heat radiated (rate)

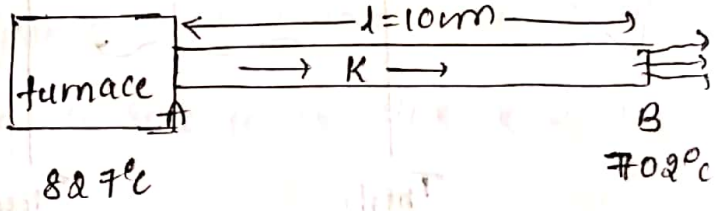
$$\sigma A T_h^4 + \sigma A T_2^4 = 2\sigma A T_1^4$$

$$T_h^4 + T_2^4 = 2T_1^4$$

$$\Rightarrow T_h^4 - T_1^4 = T_1^4 - T_2^4$$

Ex: One end A of a metallic rod is of length 10cm is inserted in a furnace whose temp is 827°C. The curved surface of the rod is insulated. The room temp is 27°C when steady state is attained, the temp of the other end B of the rod is 702°C. Find the thermal conductivity of the metal.

Sol:



In steady state,

$$\frac{KA(T_A - T_B)}{l} = \sigma A (T_B^4 - T_0^4)$$

$$K = \frac{\sigma l (T_B^4 - T_0)^4}{T_A - T_B}$$

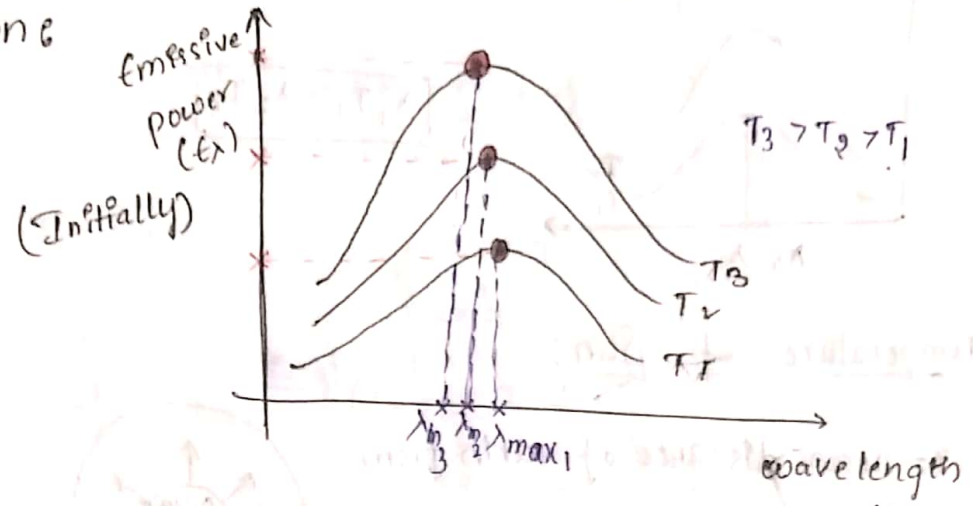
$$= \frac{5.67 \times 10 \times 10^{-11} ((975)^4 - (300)^4) \times 10^{-8}}{125}$$

$$= \frac{567}{125} ((975)^4 - (300)^4) \times 10^{-3} \times 10^{-8}$$

=

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Spectral distribution of energy in Black body radiation :



* Area under the curve gives total emissive power of the black body at that temp.

* Initially energy carried out increases with λ and get to λ_{max} & then decreases.

* As temp increases, λ_{max} shift towards left.
  └ wavelength corresponding to max intensity

Weins formula :

$$\lambda_{max} \propto \frac{1}{T}$$

$$\lambda_{max} T = \text{const} = b$$

Weins const $= b = 2.89 \times 10^{-3}$ mK

$$\boxed{\lambda_1 T_1 = \lambda_2 T_2}$$

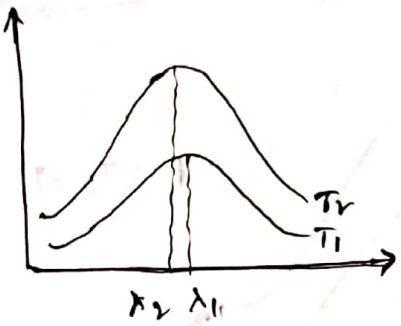
* Total emissive power, $\boxed{E \propto T^4}$

* Spectral emissive power corresponding to max intensity, $\boxed{E_{max} \propto T^5}$

$$\Rightarrow \boxed{E_{max} \propto \frac{1}{\lambda_{max}^5}}$$

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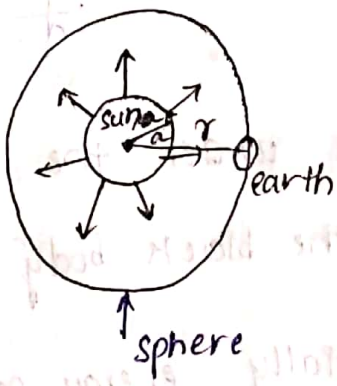
*



$$\lambda_1 T_1 = \lambda_2 T_2$$

Temperature of Sun:

r = mean distance of earth from sun



Radiating power of sun;

$P = \sigma A T_S^4 \quad (e = 1)$

$P = \sigma (4\pi a^2) T_S^4 \quad -①$

Solar constant (S) = Intensity of solar radiation at earth surface.

$$= \frac{\text{Power}}{\text{Area}} = \frac{P}{4\pi r^2} = \frac{\sigma \times 4\pi a^2 T_S^4}{4\pi r^2}$$

$$S = \frac{\sigma a^2 T_S^4}{r^2}$$

$$\boxed{T_S = \left(\frac{S r^2}{\sigma a^2}\right)^{1/4}}$$

$T_S = 6000K$.

Newton's Law of Cooling:

* From stephen's law,

rate of loss of heat, $= \frac{dQ}{dt} = \sigma e A (T^4 - T_0^4)$

rate of colling/fall in temp, ~~$\frac{dT}{dt}$~~

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$$-ms\frac{dT}{dt} = \sigma e A(T^4 - T_0^4)$$

$$\frac{dT}{dt} = -\frac{\sigma e A(T^4 - T_0^4)}{ms}$$

let $T = T_0 + \Delta T \Rightarrow \Delta T = T - T_0$ (small)

now, $$\frac{dT}{dt} = -\frac{\sigma e A}{ms}\left[(T_0+\Delta T)^4 - T_0^4\right]$$

$$= -\frac{\sigma e A}{ms}\left[T_0^4\left(1+\frac{\Delta T}{T_0}\right)^4 - T_0^4\right]$$

$$= -\frac{\sigma e A}{ms} T_0^4\left[\left(1+\frac{\Delta T}{T_0}\right)^4 - 1\right]$$

$$= -\frac{\sigma e A}{ms} T_0^4\left[1 + \frac{4\Delta T}{T_0} - 1\right]$$

$$= -\frac{\sigma e A T_0^4}{ms}\left[\frac{4\Delta T}{T_0}\right]$$

$$= \frac{-4\sigma e A T_0^3 \Delta T}{ms}$$

$$= -\frac{b}{ms}(\Delta T) \qquad \boxed{b = 4\sigma e A T_0^3}$$

$$\frac{dT}{dt} = \frac{-b}{ms}(T - T_0)$$

(or)

$$\frac{dT}{dt} = -K(T - T_0) \qquad -①$$

$K = \frac{b}{ms}$ = another cooling const

now, $$\frac{dT}{T - T_0} = -K\,dt$$

If the body cools from T_1 to T_2 (^oc) in time 't' then

$$\int_{T_1}^{T_2} \frac{dT}{T - T_0} = -K\int_0^t dt$$

$$\ln\left(\frac{T_2 - T_0}{T_1 - T_0}\right) = -Kt$$

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$$\therefore \boxed{t = \frac{1}{K} \ln\left(\frac{T_1 - T_0}{T_2 - T_0}\right)} \quad **$$

(or) $$\ln\left(\frac{T_2 - T_0}{T_1 - T_0}\right) = -Kt$$

$$\frac{T_2 - T_0}{T_1 - T_0} = e^{-Kt}$$

$$\boxed{T_2 - T_0 = (T_1 - T_0)\, e^{-Kt}} \quad *$$

(or) from eqn ①

$$\frac{dT}{dt} = -K(T - T_0)$$

$$\boxed{\frac{T_1 - T_2}{t} = -K\left(\frac{T_1 + T_2}{2} - T\right)}$$

Avg form of Newton's Law of cooling.

Cooling curves (or) Graphs :

① $\ln(T - T_0)$ Vs time graphs :-

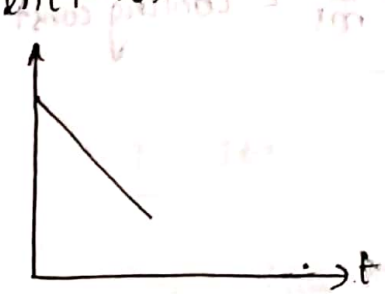
we have, $$\int_{T_1}^{T} \frac{dT}{T - T_0} = -K \int_0^t dt$$

$$\ln\left(\frac{T_1 - T_0}{T_2 - T_0}\right) = Kt$$

$$\ln(T_1 - T_0) - \ln(T - T_0) = Kt$$

$$\ln(T - T_0) = -Kt + \ln(T_1 - T_0)$$

$$(y = mx + c)$$



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② Temp of body Vs time :

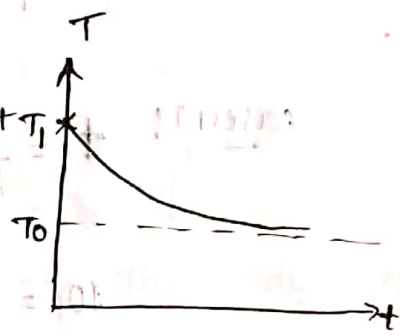
Again we have,

$$\ln\left(\frac{T-T_0}{T_1-T_0}\right) = -Kt$$

$$\frac{T-T_0}{T_1-T_0} = e^{-Kt}$$

$$T-T_0 = (T_1-T_0)e^{-Kt}$$

$$T = T_0 + (T_1-T_0)e^{-Kt}$$



when $t=0 \Rightarrow T = T_1$

when $t=\infty \Rightarrow T = T_0$

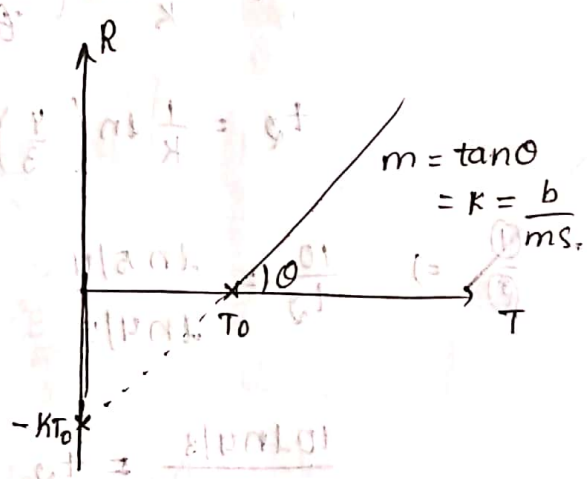
③ Rate of cooling Vs temp of body :

we have,

$$-\frac{dT}{dt} = R = K(T-T_0)$$

$$R = KT - KT_0$$

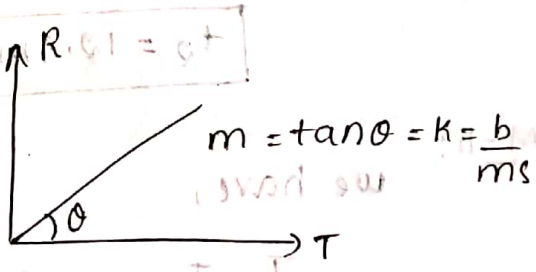
$$(y = mx + c)$$



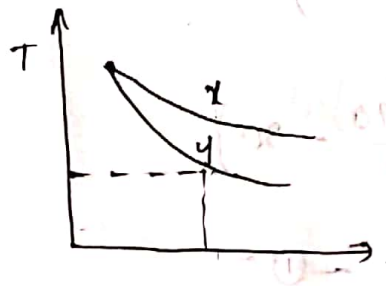
④ Rate of cooling Vs temp diff :

$$-\frac{dT}{dt} = R = k(T-T_0)$$

$$(y = mx)$$



⑤



$e_y > e_x$

$a_y > a_x$

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Ex: A body cools from 80°c to 70°c in 10min. Find the time required further for it to cool from 70°c to 60°c. Assume temp of surrounding to be 30°c.

Sol:

M-I:

we have,

$$t = \frac{1}{K} \ln\left(\frac{T_1 - T_0}{T_2 - T_0}\right)$$

case(1): $t_1 = \frac{1}{K} \ln\left(\frac{80-30}{70-30}\right)$

$$10 = \frac{1}{K} \ln\left(\frac{5}{4}\right) \quad -(1)$$

case(2): $t_2 = \frac{1}{K} \ln\left(\frac{70-30}{60-30}\right)$

$$t_2 = \frac{1}{K} \ln\left(\frac{4}{3}\right) \quad -(2)$$

$\frac{(1)}{(2)} \Rightarrow \frac{10}{t_2} = \frac{\ln 5/4}{\ln 4/3}$

$$\frac{10 \ln 4/3}{\ln 5/4} = t_2$$

$$\boxed{t_2 = 12.9 \text{ min}}$$

M-II:

we have,

$$\frac{T_1 - T_2}{t} = -K\left(\frac{T_1 + T_2}{2} - T_0\right)$$

case(1): $\frac{80-70}{10} = -K\left(\frac{80+70}{2} - 30\right)$

$$1 = -K(45) \quad -(1)$$

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case (2):

$$\frac{70-60}{t_2} = -k\left(\frac{70+60}{2} - 30\right)$$

$$\frac{10}{t_2} = -k(35)$$

$$\frac{10}{t_2} = \frac{1}{45} \times 7$$

$$t_2 = \frac{90}{7} =$$

Ex: Assume surroundings to be 25°C. The body takes 5 min to cool from 45°C to 40°C. Find the temp after further 5 min.

Sol:

M-I:

$$t = \frac{1}{k} \ln\left(\frac{T_1 - T_0}{T_2 - T_0}\right)$$

case (1):

$$5 = \frac{1}{k} \ln\left(\frac{20}{15}\right) = \frac{1}{k} \ln\left(\frac{4}{3}\right)$$

$$5 = \frac{1}{k} \ln(4/3) \quad \text{— (1)}$$

case (2):

$$5 = \frac{1}{k} \ln\left(\frac{40-25}{T_2 - 25}\right) \quad \text{— (2)}$$

(1) & (2) $\Rightarrow$

$$\frac{\ln(4/3)}{\ln(15/T_2 - 25)}$$

$$\ln\left(\frac{15}{T_2 - 25}\right) = 2\ln\left(\frac{4}{3}\right)$$

$$\frac{15}{T_2 - 25} = \frac{4}{3}$$

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$$45 = 4T_2 - 100$$

$$4T_2 = 145$$

$$\boxed{T_2 = 36.25}$$

M-II:

$$\frac{T_1 - T_2}{t} = -K\left(\frac{T_1 + T_2}{2} - T_0\right)$$

case(1): $$\frac{5}{5} = -K\left(\frac{85 - 50}{2}\right)$$

$$1 = -K\left(\frac{35}{2}\right) \quad -(1)$$

case(2):

$$\frac{40 - T_2}{5} = -K\left(\frac{40 + T_2 - 50}{2}\right)$$

$$\frac{40 - T_2}{5} = \frac{2}{35}\left(\frac{T_2 - 10}{2}\right)$$

$$280 - 7T_2 = T_2 - 10$$

$$290 = 8T_2$$

$$T_2 = \frac{145}{4}$$

$$\boxed{T_2 = 36.25}$$

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Ex! A calorimeter contains 50gms of water at 50°c. The temp falls to 45°c in 10mins. when the calorimeter contains 100gms water at 50°c. It takes 18 mins for the temp to fall to 45°c. Calculate water equivalent of calorimeter.

Sol!

M-I:

$$T_2 - T_0 = (T_1 - T_0)\, e^{\frac{-bt}{ms}}$$

w = water equivalent of calorimeter

case(1)!

$$(45 - T_0) = (50 - T_0)\, e^{\frac{-b\times 10}{ms = w+50}} \quad \text{— (1)}$$

case(2)!

$$(45 - T_0) = (50 - T_0)\, e^{\frac{-b\times 18}{w+100}} \quad \text{— (2)}$$

$$\frac{5}{\frac{-10b}{w+50}} = \frac{-b\times 18}{w+100}$$

$$5w + 500 = 9w + 450$$

$$50 = 4w$$

$$\boxed{w = 12.5 gms}$$

M-II:

Average Method,

$$\frac{T_1 - T_0}{t} = -k\left(\frac{T_1 + T_2}{2} - T_0\right)$$

$$\frac{T_1 - T_0}{t} = -\frac{b}{ms}\left(\frac{T_1 + T_2}{2} - T_0\right)$$

where $\left(k = \frac{b}{ms}\right)$.

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case(1):

$$\frac{50-T_0}{10} = -\frac{b}{\omega+50}\left(\frac{50+45}{2}-T_0\right) \quad -(1)$$

case(2):

$$\frac{50-T_0}{18} = -\frac{b}{\omega+100}\left(\frac{50+45}{2}-T_0\right) \quad -(2)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{\frac{1}{10}5}{\frac{1}{18}9} = \frac{\frac{-b}{\omega+50}}{\frac{-b}{\omega+100}}$$

$$\frac{9}{5} = \frac{\omega+100}{\omega+50}$$

$$\Rightarrow \boxed{\omega = 12.5 \text{ gms}}$$

Ex: A calorimeter of water equivalent 100 gms cools in air in 18 min from 60°c to 40°c. when a block of metal of mass 80gms is heated to 60°c & placed inside the calorimeter. Assume heat loss only by radiation & Newton's law of cooling is valid. Find specific heat of metal, If now the system cools from 60°c to 40°c in 20 min.

Sol:

M-I:

$$T_g - T_0 = (T_1 - T_0)\, e^{-\frac{bt}{ms}}$$

case(1):

$$40 - T_0 = (60 - T_0)\, e^{-\frac{b \times 18}{100}}$$

case(2):

$$40 - T_0 = (60 - T_0)\, e^{-\frac{b \times 20}{60s+100}}$$

$$\frac{18}{100} = \frac{20}{60s+100}$$

$$s = \frac{10}{54}$$

$$\boxed{s = 0.185}$$

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M-II:

$$\frac{T_1 - T_0}{t} = \frac{-b}{ms}\left(\frac{T_1 + T_2}{2} - T_0\right)$$

case(i):

$$\frac{60 - T_0}{18} = \frac{-b}{100}\left(\frac{60+40}{2} - T_0\right) \quad -(1)$$

$$\frac{60 - T_0}{20} = \frac{-b}{60s+100}\left(\frac{60+40}{2} - T_0\right) \quad -(2)$$

$\frac{(1)}{(2)} \Rightarrow$ [illegible]

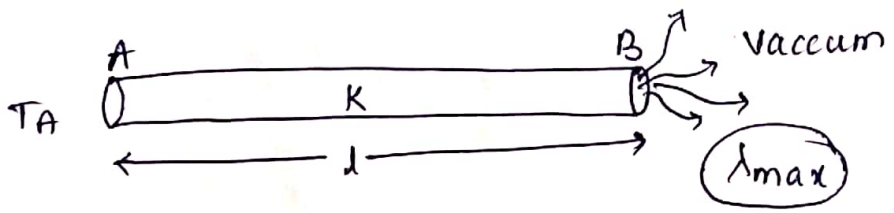
$$\frac{18b}{5} = \frac{20b}{60s+10} \Rightarrow \frac{9}{5} = \frac{20}{60s+10} \Rightarrow \boxed{s = 0.185}$$

Ex: The maximum in the energy distribution spectrum of the sun is at 4753 A° units, and it's temp is 6050K. what will be the temp of the star whose energy distribution shows a max of 9506A° units.

$$\lambda_1 T_1 = \lambda_2 T_2$$

$$4753 \times 6050 = 9506 \times T_2$$

$$\boxed{T_2 = 3025K}$$



$$\frac{KA(T_A - T_B)}{l} = \sigma A T_B^4$$

$$\lambda_m T_B = b$$

$$T_B = \frac{b}{\lambda_{max}}$$

$$\frac{K(T_A - T_B)}{l} = \sigma T_B^4$$

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